

Emerging Technologies for the Circular Economy

Lecture 4: Lifecycle Assessment (LCA) - Part 2

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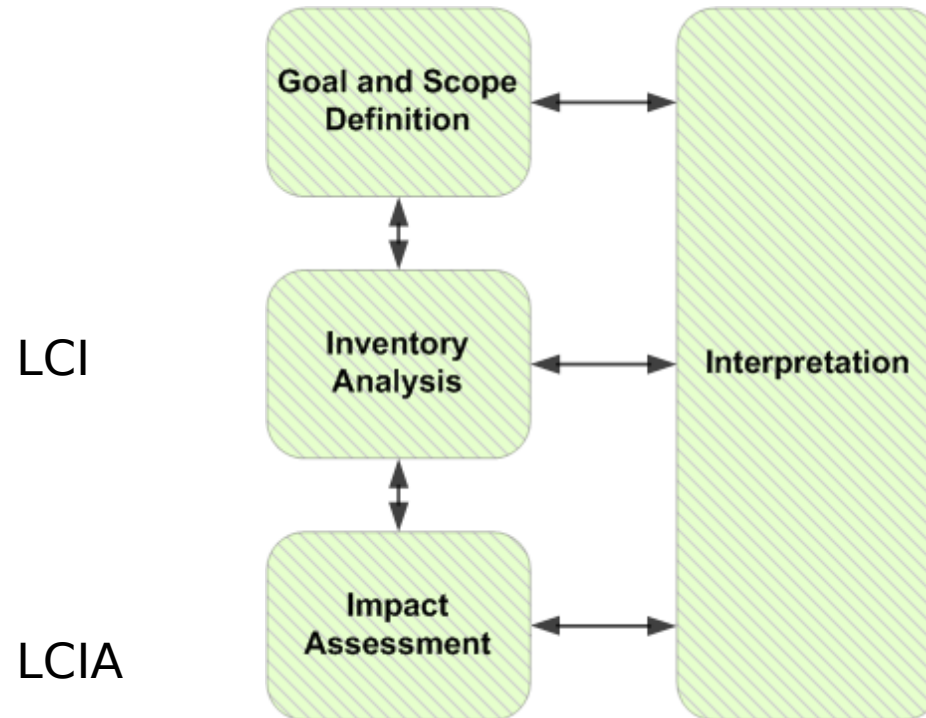
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- Updated versions of these slides will be available in our [Github repository](#).

RECAP

Lifecycle Assessment (LCA)

The four main stages



Goal and Scope Definition

Goal of an LCA study

- **ISO 14040 definition**
- The goal of an LCA states:
 - The intended application
 - The reasons for carrying out the study
 - The intended audience
 - Whether the results are intended to be used in comparative assertions released publicly

ISO 14040 Environmental management — Life cycle assessment — Principles and framework, International standards organisation (<https://www.iso.org/standard/37456.html>)

Determining the environmental impacts of conventional and alternatively fuelled vehicles through LCA, Ricardo Energy and Environment ([Link](#))

Goal and Scope Definition

Scope of an LCA study

- **ISO 14040 standard**
- The scope of an LCA should describe:
 - **The functional unit(s) of the system(s)**
 - **Reference flow(s)**
 - The system boundary
 - LCIA methodology and types of impacts analysed
 - *Limitations*
 - *Data quality requirements*
 - ...

quantified performance of a product system for use as a reference unit, e.g 1 million holes drilled

measure of the product(s) or product parts required to deliver the performance defined by the functional unit.

Goal and Scope Definition

Scope of an LCA study

- Dishwasher



“View of dishwasher inside”, authored by “Ryan from Louisville, USA”, licensed under [Creative Commons Attribution-Share Alike 2.0](#)

“Unwashed dishes in a sink; an authentic situation”, authored by Mysid, Public Domain

- Handwash



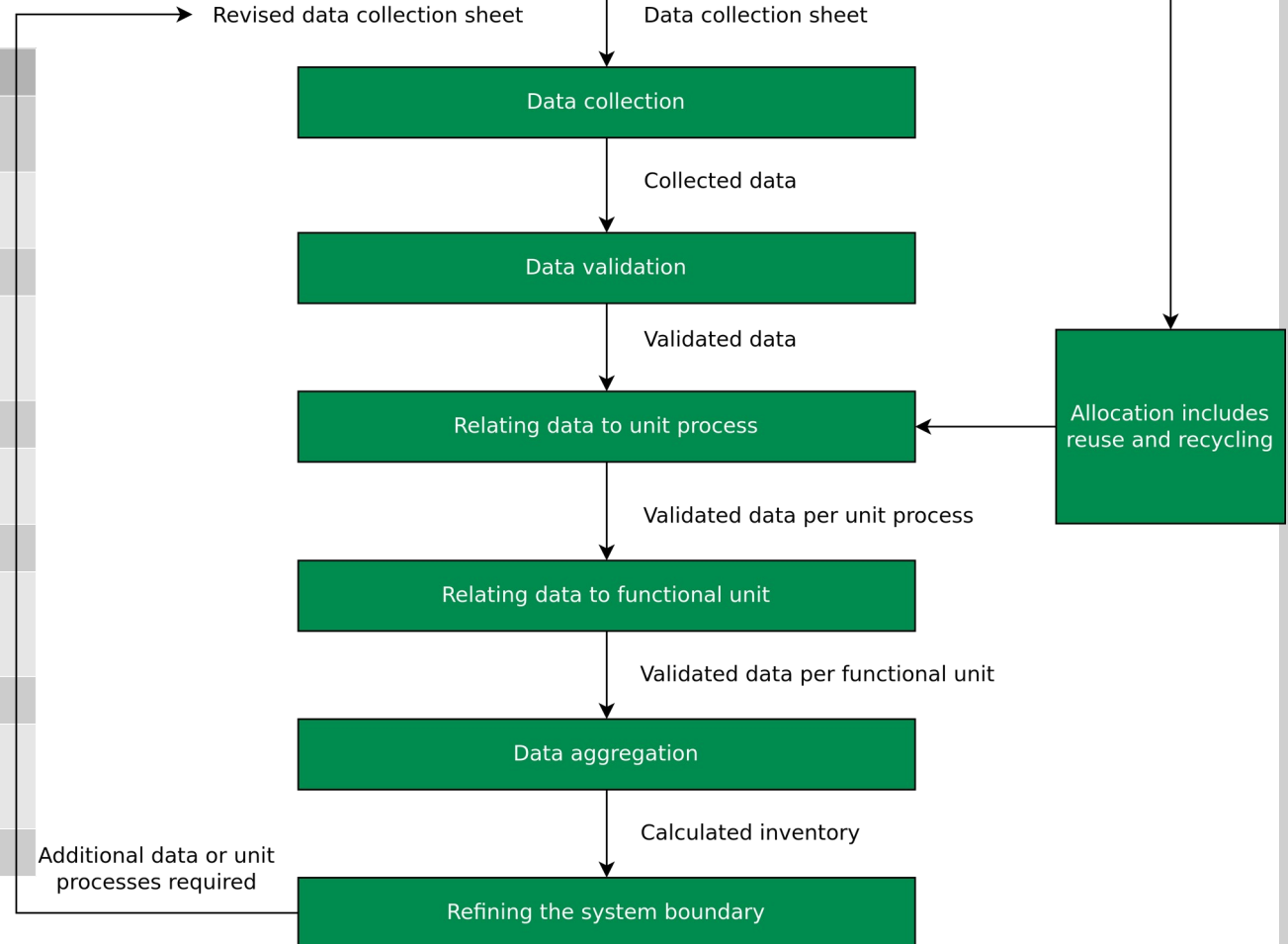
Vs.

Which is better?

Lifecycle Inventory Analysis (LCI)

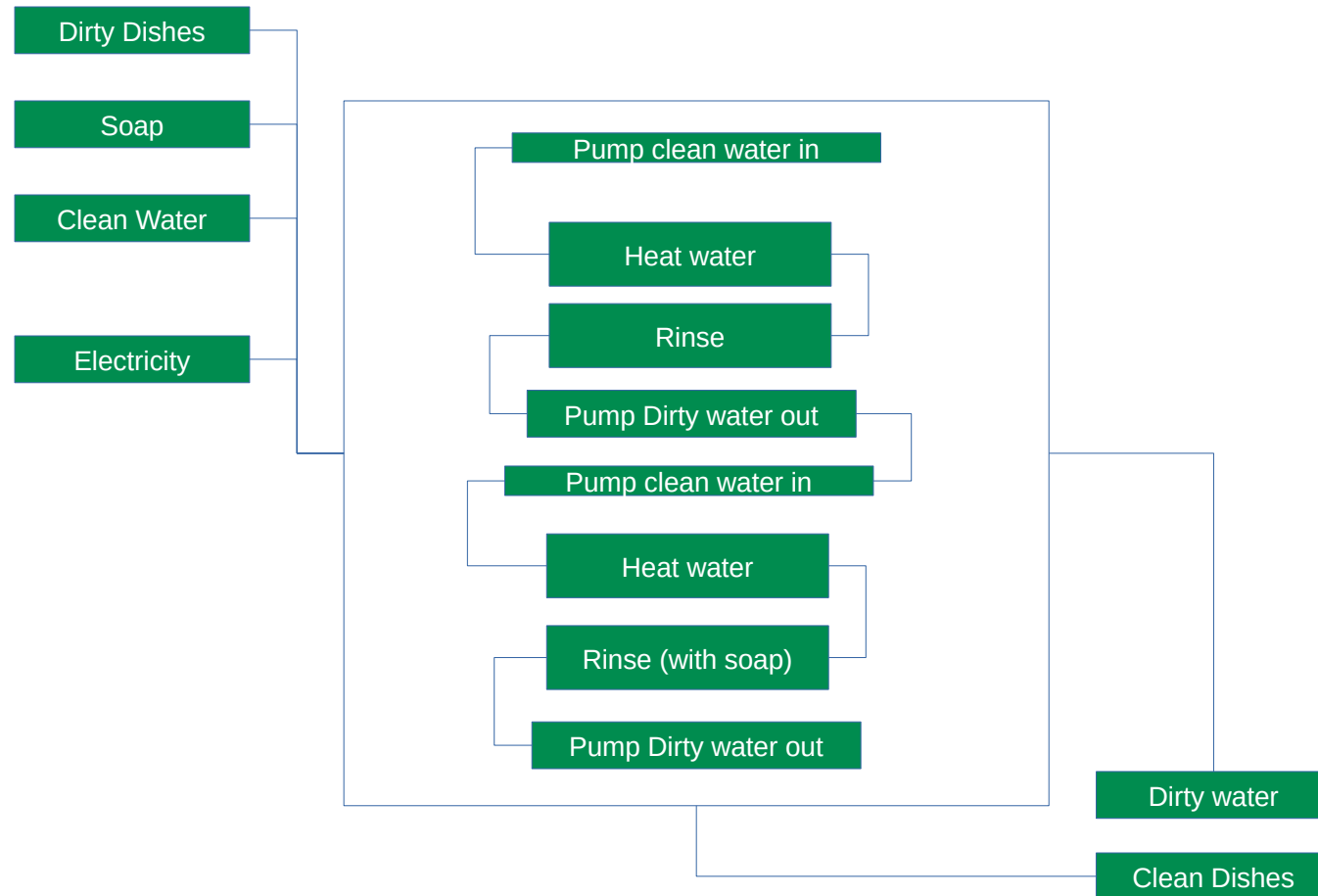
Preparing for data collection

Completed by:	Date of completion:			
Unit process identification:	Reporting location:			
Time period: Year	Starting month:	Ending month:		
Description of unit process:				
Material inputs	Units	Quantity	Description of sampling procedures	Origin
...	...	...	...	...
Water consumption	Units	Quantity		
...	...	...	...	...
Energy Inputs	Units	Quantity	Description of sampling procedures	Origin
...	...	...	...	...
Material outputs	Units	Quantity	Description of sampling procedures	Destination
...	...	...	...	...



Lifecycle Inventory Analysis (LCI)

Relating data to unit process and functional unit



■ Dishwasher example

— Experimental estimation:

- 1 wash-cycle with 20 dishes, 20 cups and 20 cutlery items used **1.17 kWh** of energy, **12 Litres** of clean water and 30g of soap

— But the functional unit was: “2 million dishes, cups, cutlery washed”, so we simply multiply the above numbers by **1,000,000**:

- 1.17 GWh, 12 million litres, 30,000 Kg of soap + **200 dishwashers!**

LIFECYCLE IMPACT ASSESSMENT (LCIA)

Lifecycle Impact Assessment (LCIA)

Definition

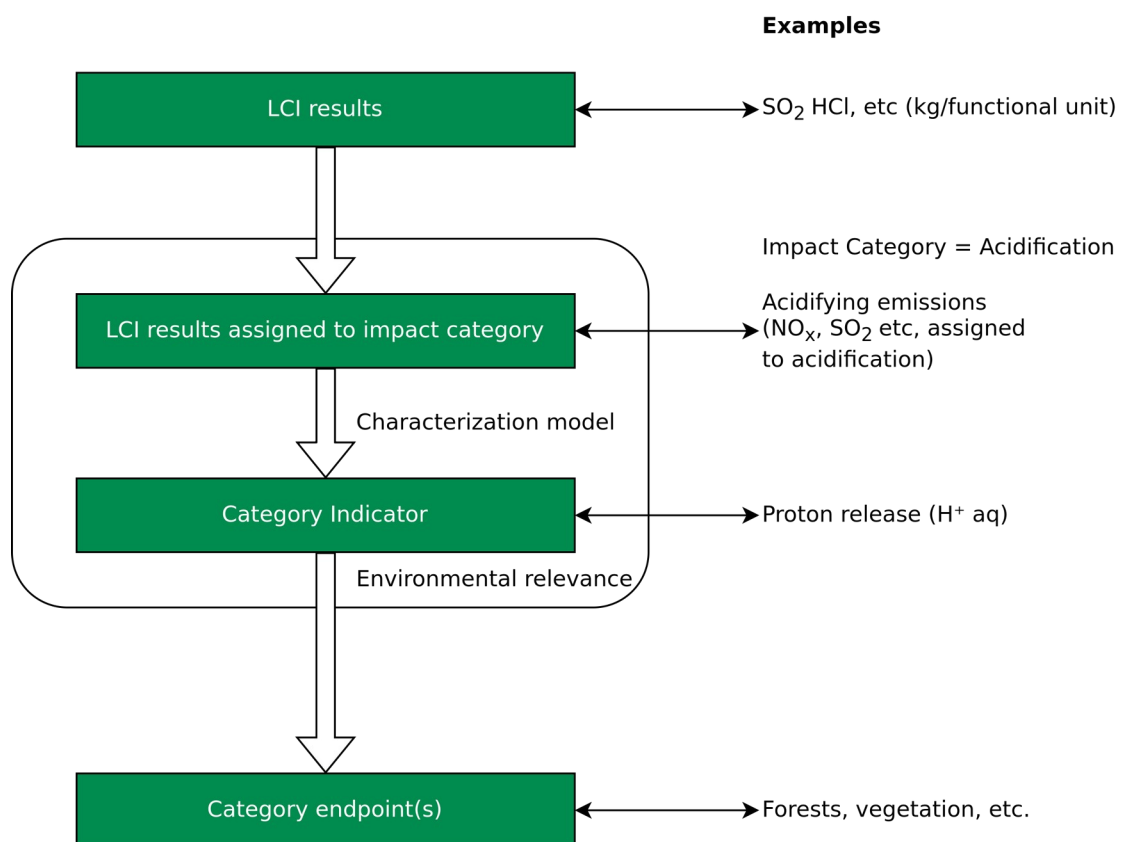
LCIA is the phase of life cycle assessment aimed at understanding and evaluating the magnitude and significance of the potential environmental impacts for a product system throughout the life cycle of the product.

ISO 14040 Environmental management — Life cycle assessment — Principles and framework, International standards organisation (<https://www.iso.org/standard/37456.html>)

Determining the environmental impacts of conventional and alternatively fuelled vehicles through LCA, Ricardo Energy and Environment ([Link](#))

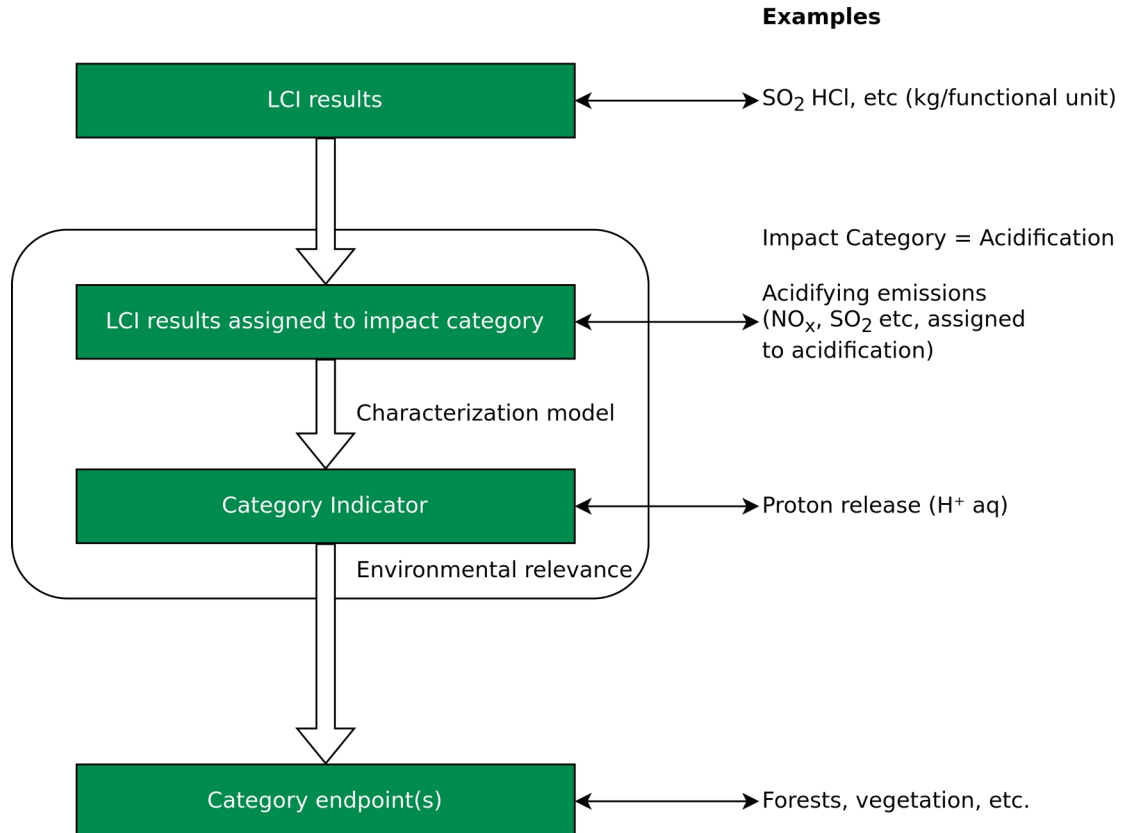
Lifecycle Impact Assessment (LCIA)

Overview



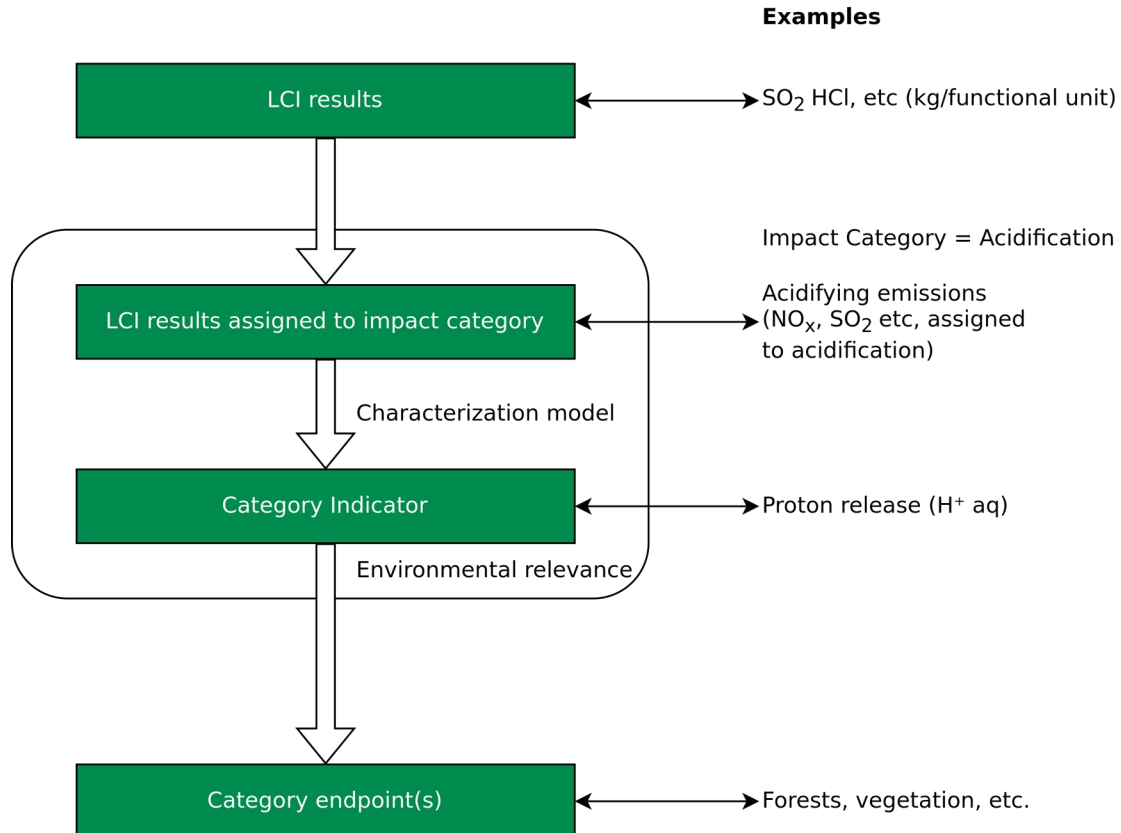
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Impact Category	Climate change
LCI results	Emission(s) per Functional Unit
Characterization model	Mathematical model that describes the environmental mechanism linking LCI results to a specific category indicator.
Category indicator	Quantifiable representation of an impact category
Charecterization factor	Factor derived from a characterization model which is applied to convert an assigned LCI result to the common unit of the category indicator
Category indicator result	Total potential impact of a product system for a specific impact category, expressed in the common unit of the selected category indicator.
Category endpoints	Qualitative or quantitative attributes of the environment, human health, or resources that the impact assessment aims to protect.
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Lifecycle Impact Assessment (LCIA) Overview



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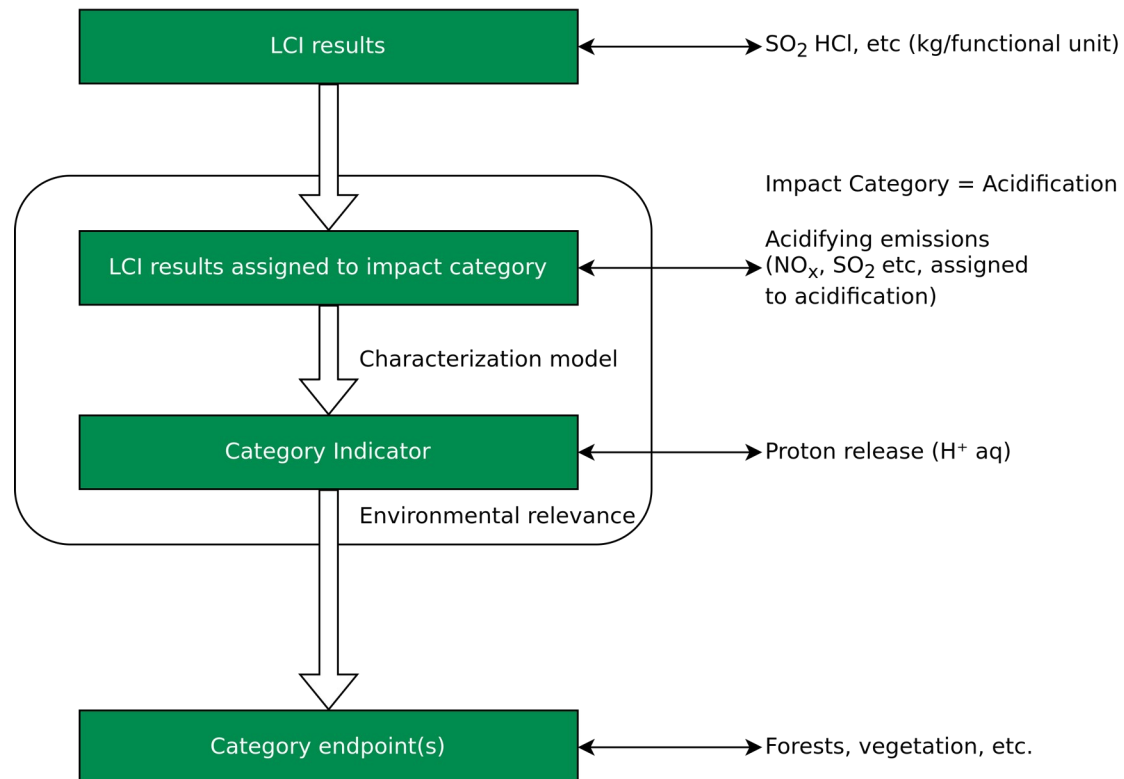
Lifecycle Impact Assessment (LCIA) Overview



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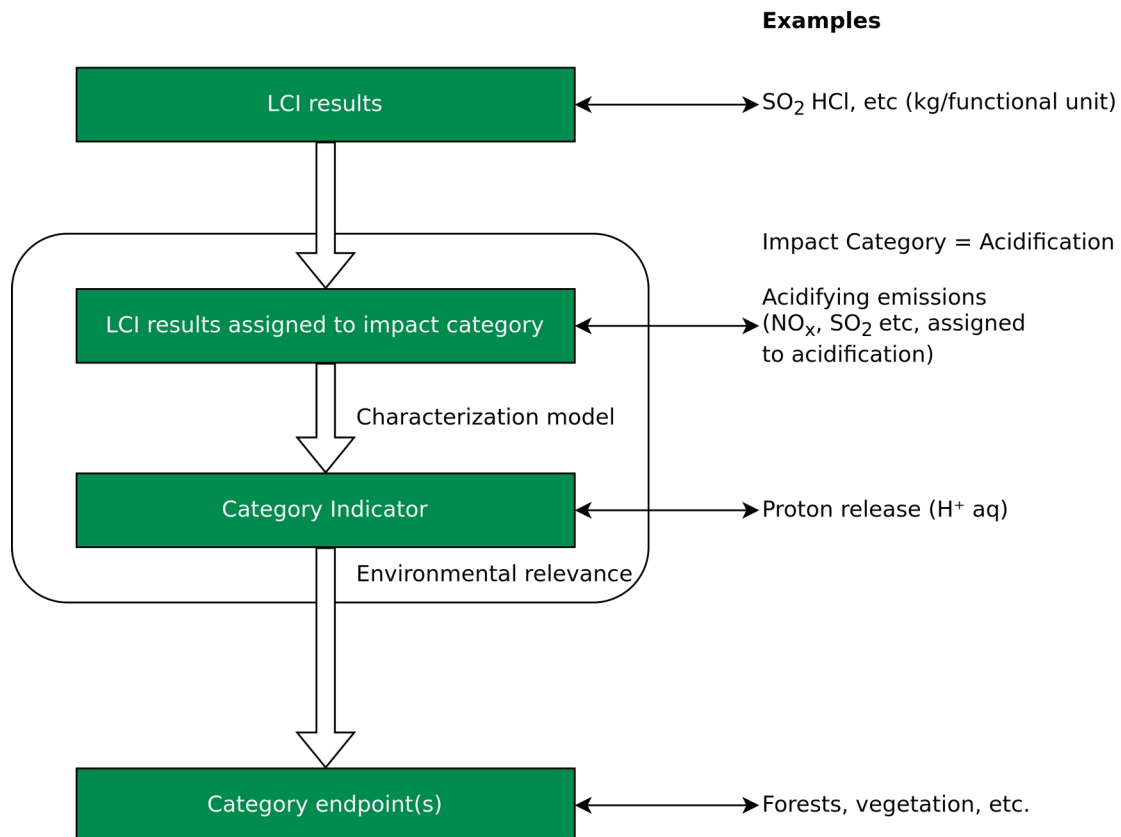
Lifecycle Impact Assessment (LCIA) Overview

Examples



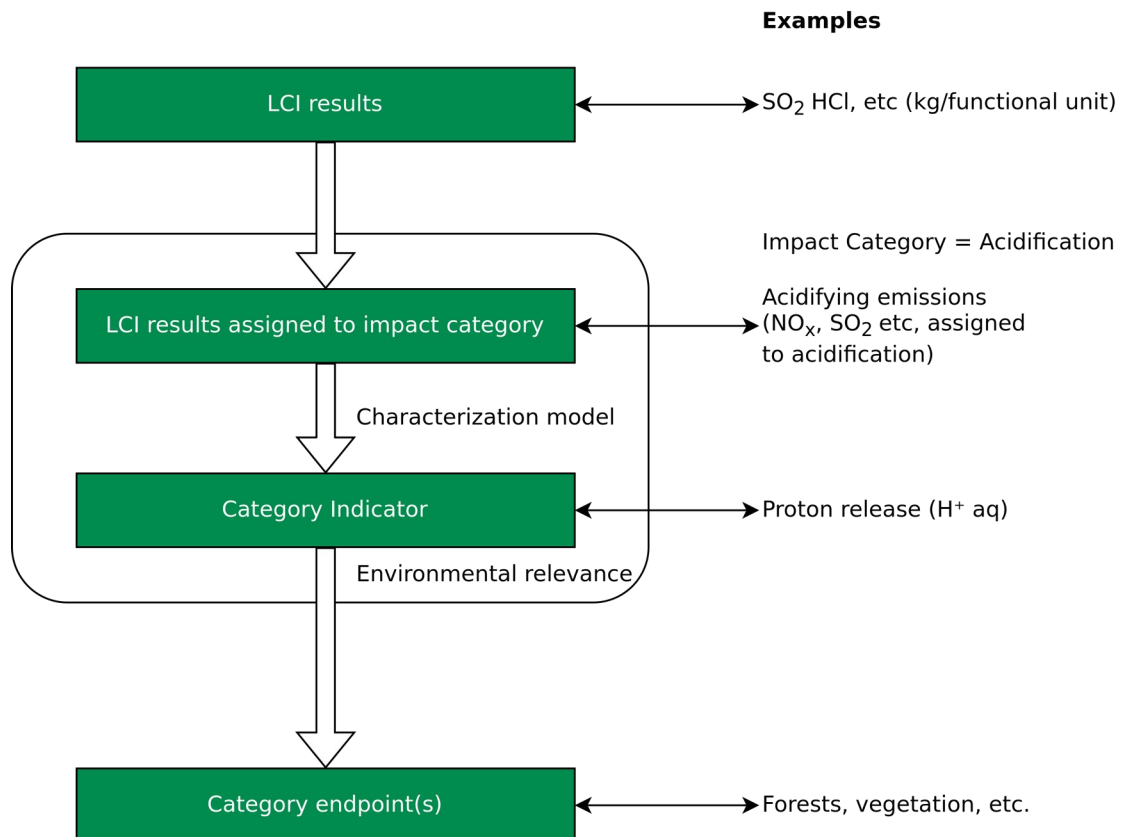
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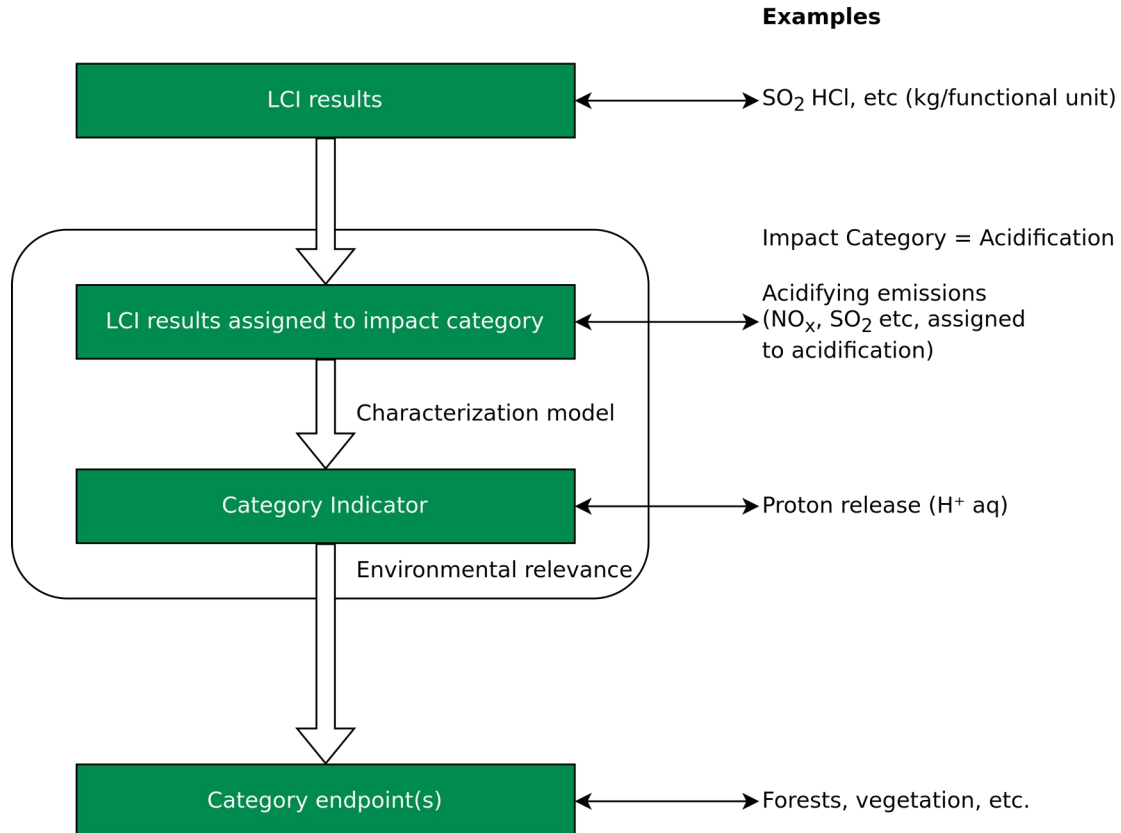
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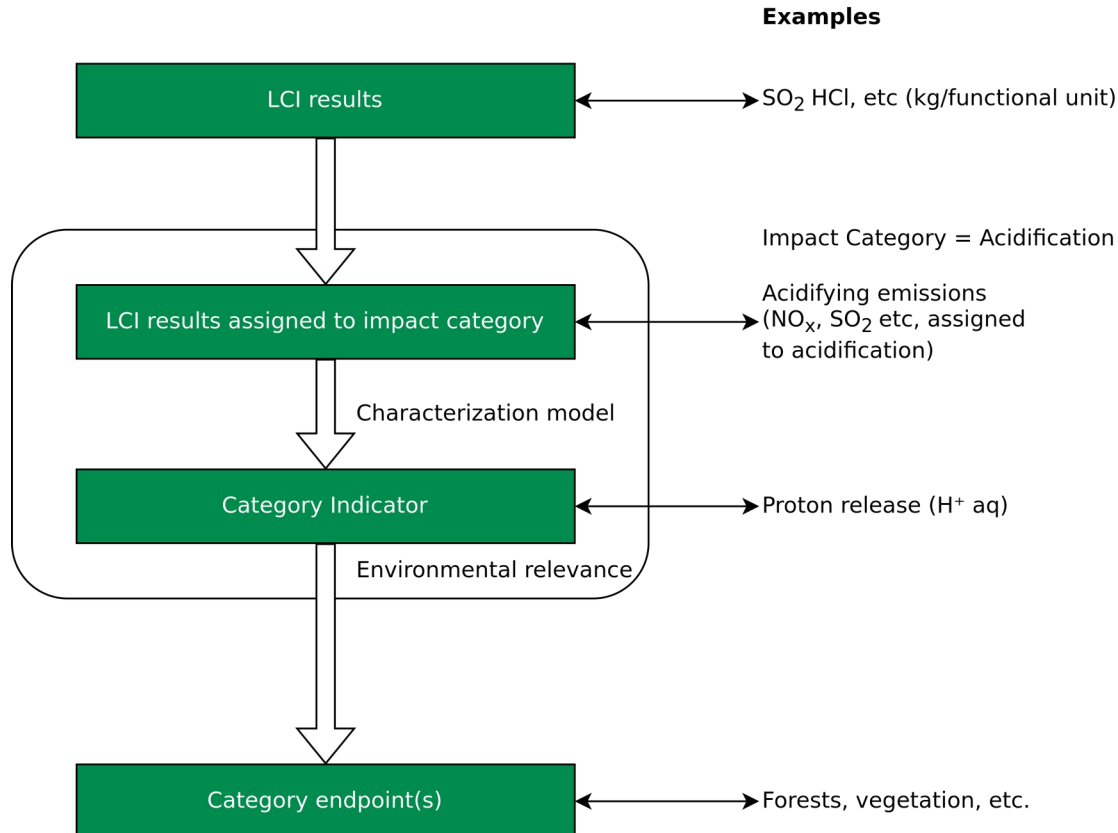
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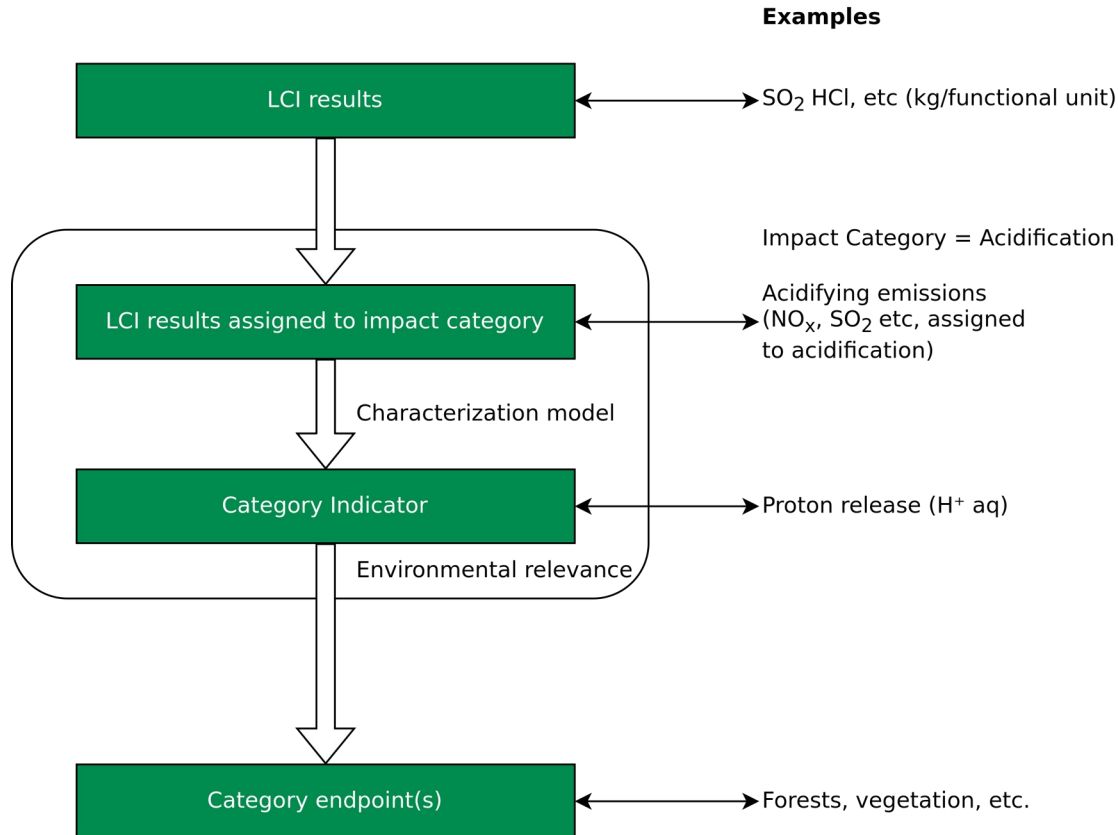
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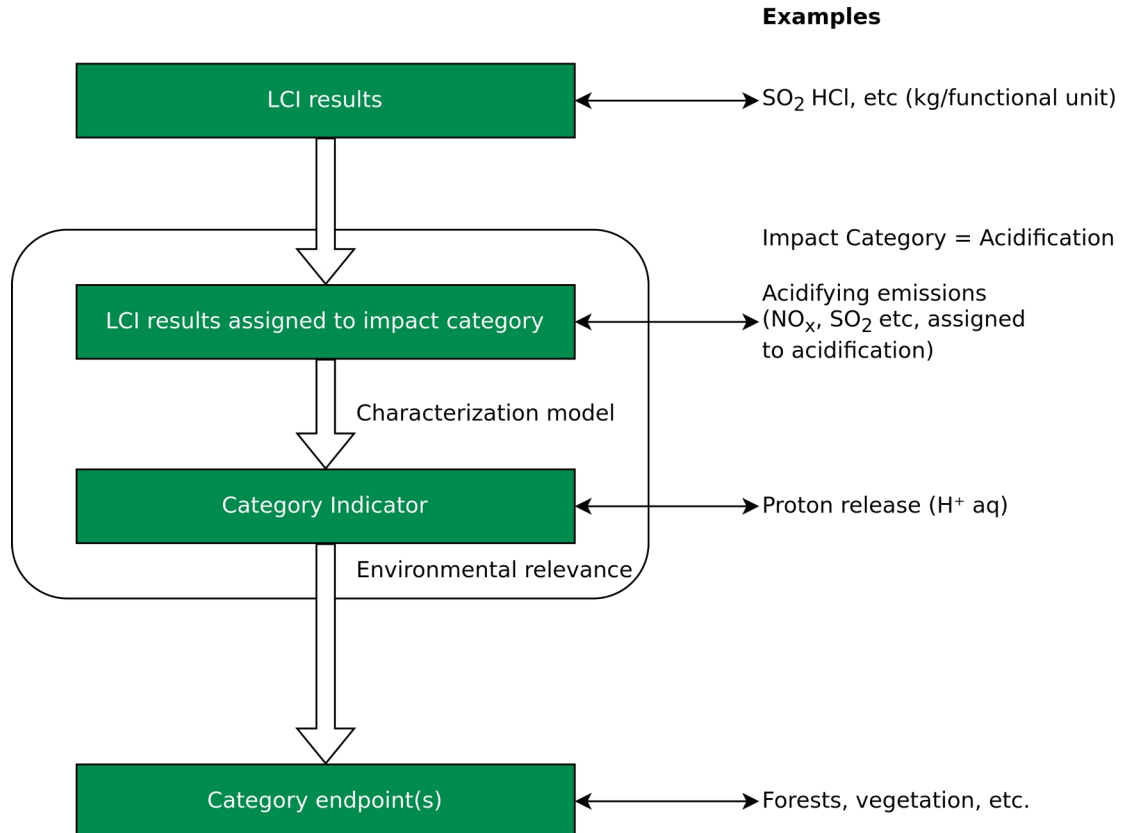
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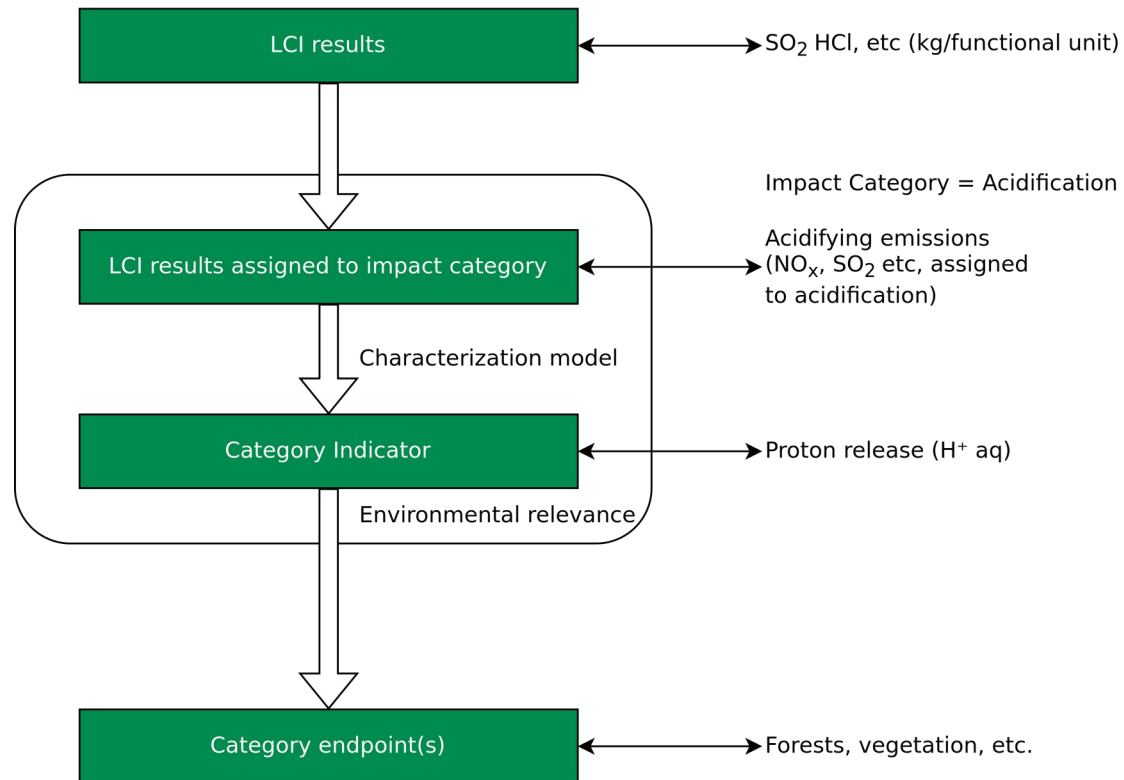
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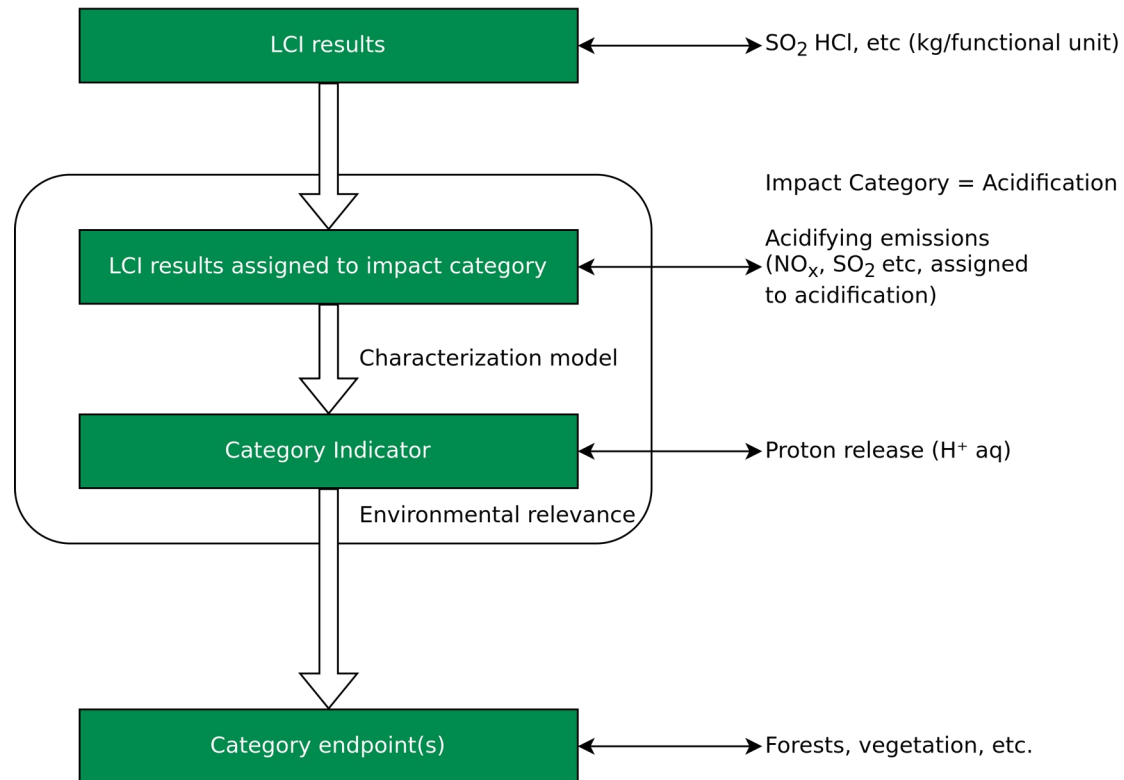
Examples



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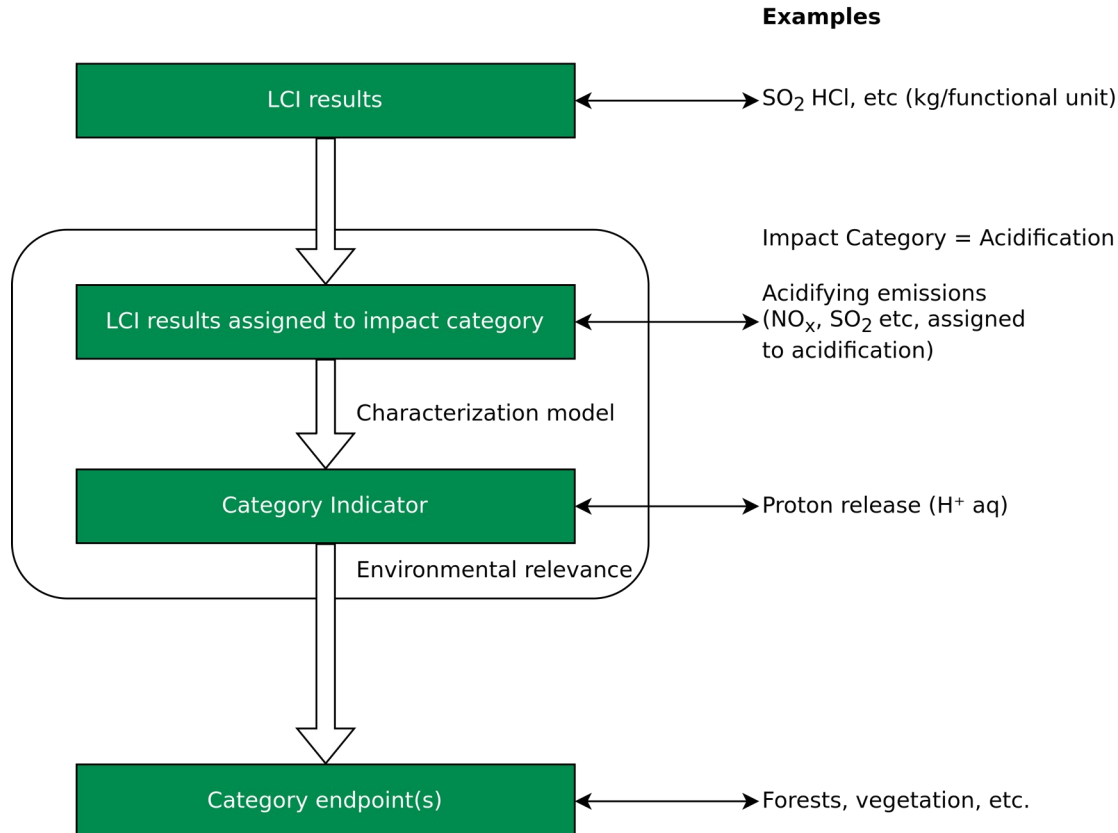


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GWP = measure of how much heat (W/m^2) a greenhouse gas traps in the atmosphere over a specific time period, relative to carbon dioxide (CO_2)

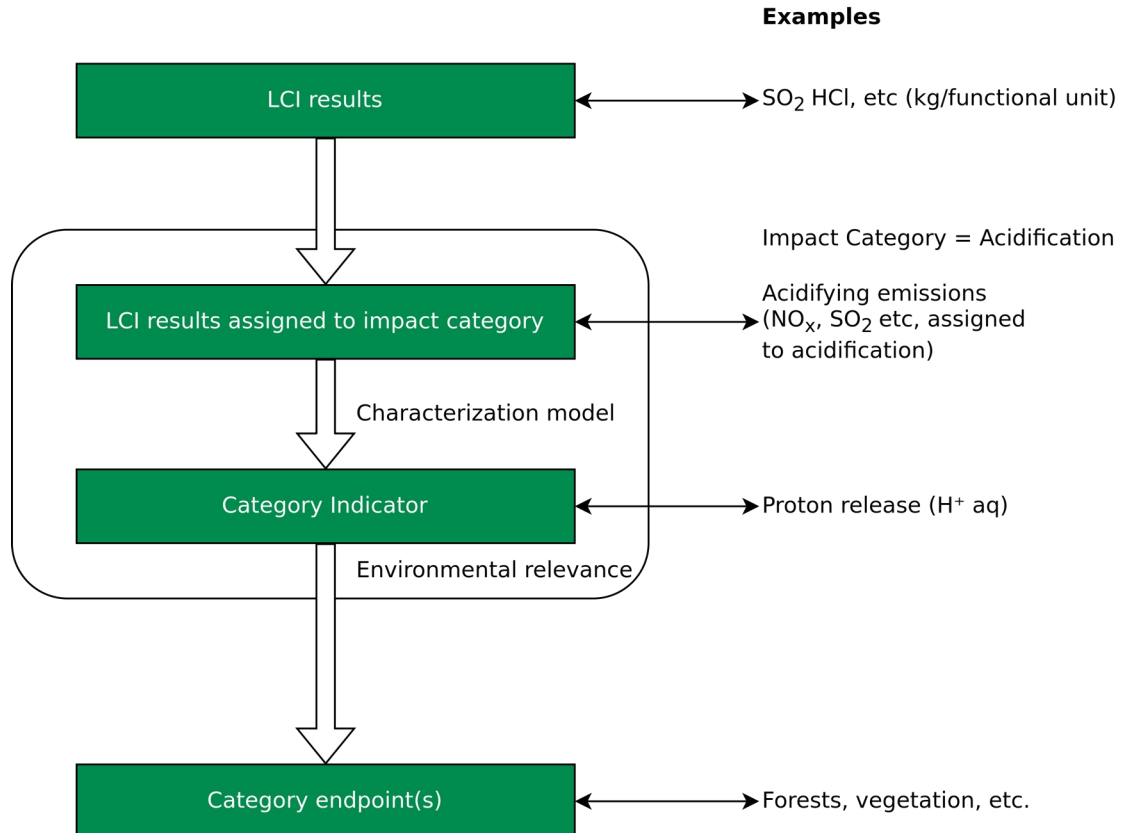
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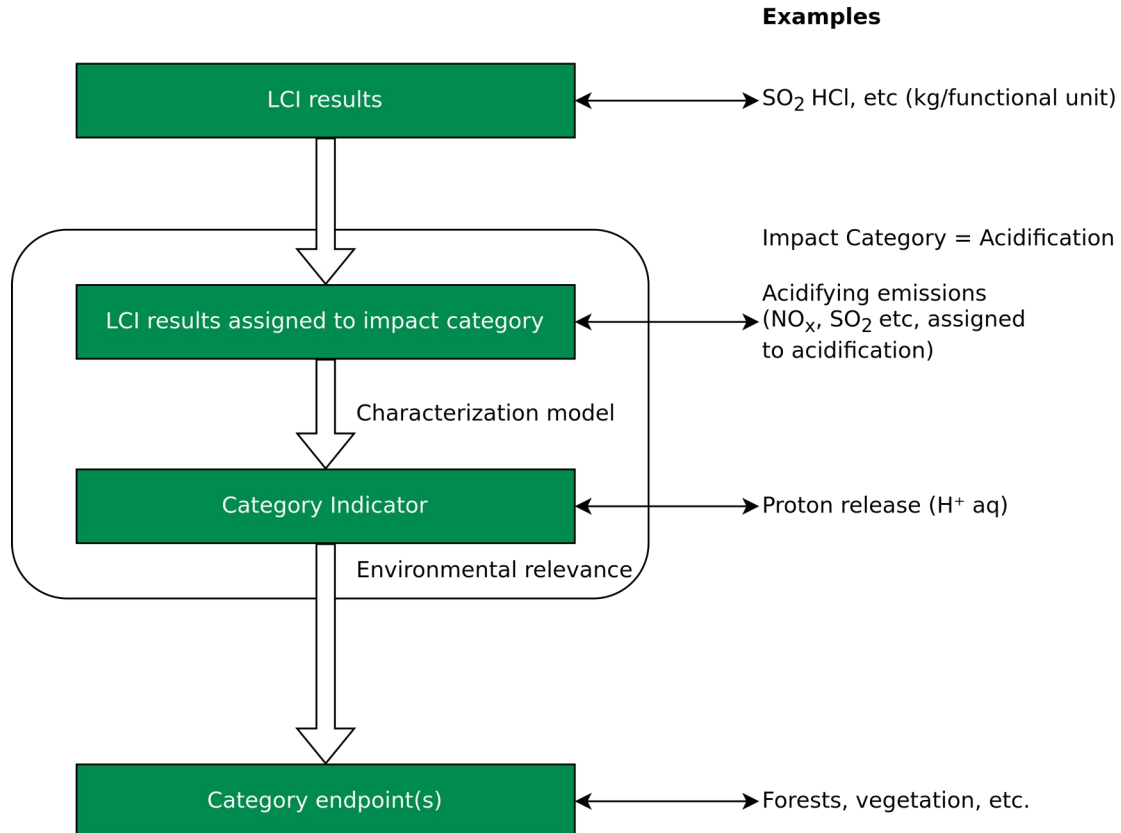
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Lifecycle Impact Assessment (LCIA)

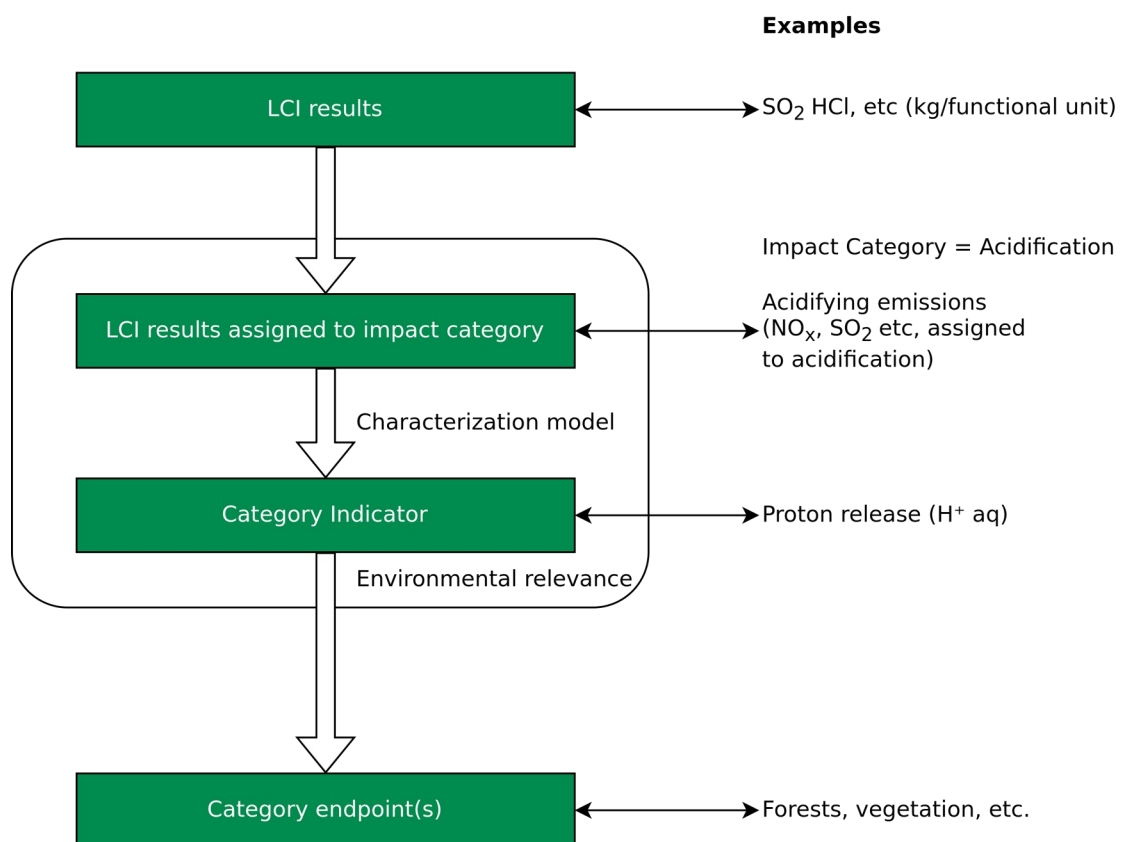
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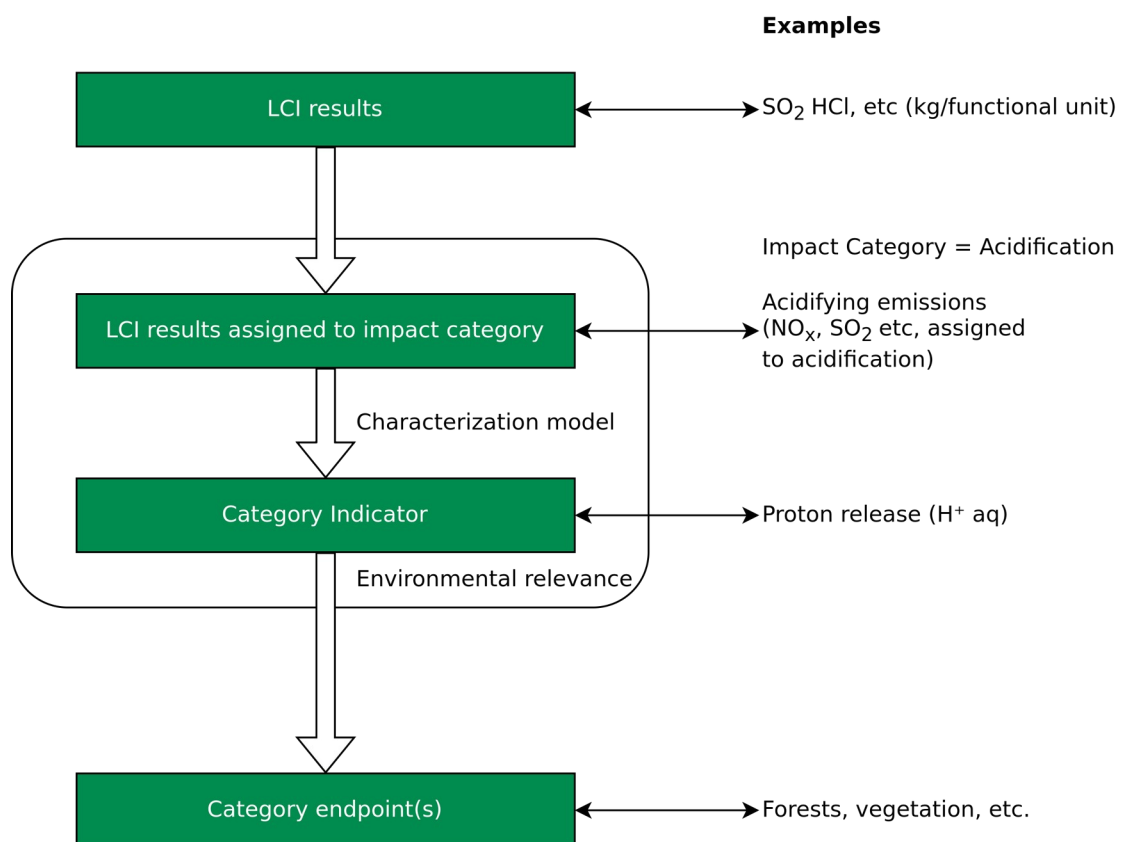
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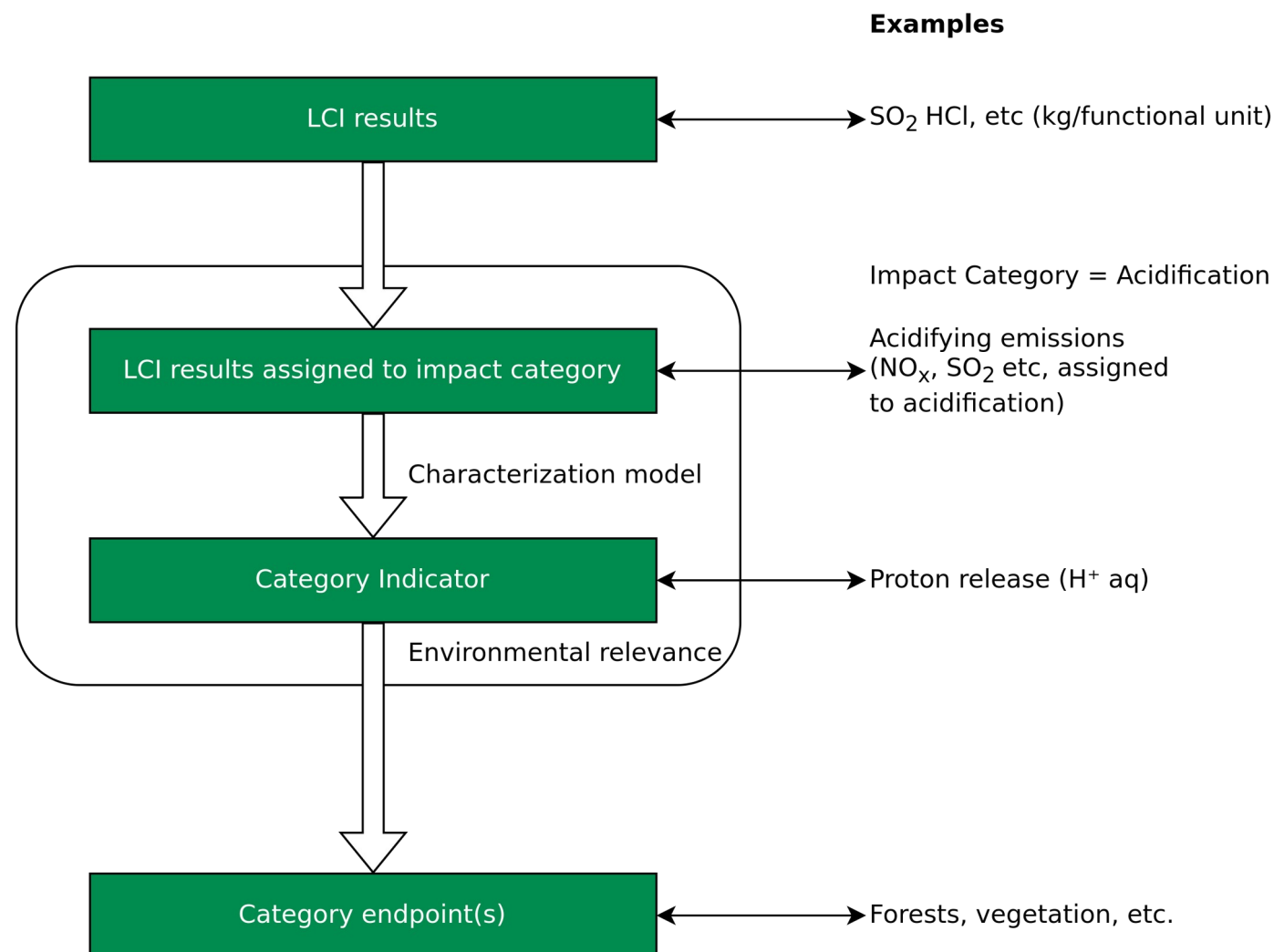


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Endpoint Indicators: Disability-Adjusted Life Years (DALY), Loss of Species per year

Lifecycle Impact Assessment (LCIA)

Overview

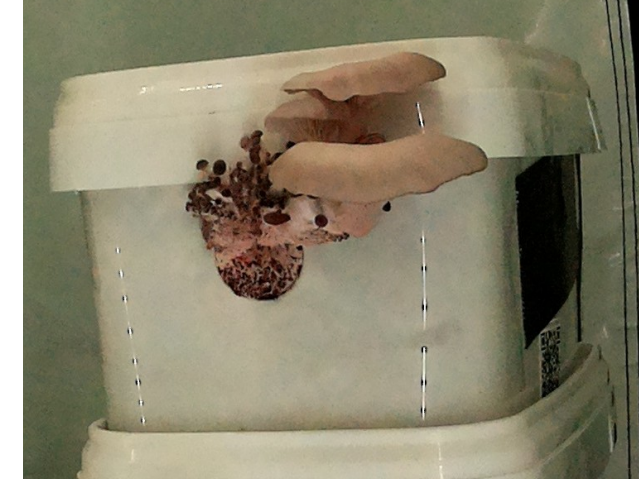


MushR: A Short Detour

LCA on Substrate Containers?

■ Goal

- Compare the environmental impact of the developed mushroom pods with non-reusable substrate bags by running lifecycle assessment calculations.



Lifecycle Inventory Analysis (LCI)

Relating data to unit process and functional unit

■ MushR example

- Experimental estimation:
 - Colonizable volume of a 5L substrate bag weighing 30g = 3L
 - *Since we have to fold the bag a couple times to seal it.*
 - Colonizable volume of a 3L substrate bucket weighing 90g = 3L (*no change*)
- This allows us to scale and compare the two container types on similar terms, e.g:
 - 1000g of substrate bags, can contain $(1000 \times 3 / 30) = 100$ Liters of substrate.
 - 1000g of substrate buckets, can contain $(1000 \times 3 / 90) = 33.333$ Liters of substrate.

Lifecycle Impact Assessment (LCIA)

MushR Example

Impact category	Reference unit	Mushroom Substrate Bag	MushR Reusable Pods
Acidification	H+ mmole eq	11.21	40.15
Global warming	g CO2 eq	102.46	217.20
Ozone depletion	g CFC-11 eq	2.64E-07	1.49E-06
Water intake	liters	0.95	2.3

Tables recreated from Determining the environmental impacts of conventional and alternatively fuelled vehicles through LCA, Ricardo Energy and Environment ([Link](#))

Lifecycle Impact Assessment (LCIA)

MushR Example

Impact category	Reference unit	Mushroom Substrate Bag	MushR Reusable Pods	MushR Pods Break-even point (reuse cycles)
Acidification	H+ mmole eq	11.21	40.15	3.6
Global warming	g CO2 eq	102.46	217.20	2.1
Ozone depletion	g CFC-11 eq	2.64E-07	1.49E-06	5.6
Water intake	liters	0.95	2.3	2.4

Tables recreated from Determining the environmental impacts of conventional and alternatively fuelled vehicles through LCA, Ricardo Energy and Environment ([Link](#))



Lifecycle Impact Assessment (LCIA)

2020 EU Study Example

Impact Category	Indicator and unit
Climate change	Greenhouse gas emissions GWP100 in CO ₂ eq (including carbon feedbacks)
Energy consumption	Cumulative energy demand in MJ: non-renewable (fossil and nuclear) and renewable
Acidification	Acidification potential in SO ₂ eq
Eutrophication	Eutrophication potential in PO ₄ ³⁻ eq
Photochemical ozone formation	Photochemical Ozone Creation Potential POCP in NMVOC eq
Ozone depletion	ODP in R11 eq
Ionising radiation	Ionising radiation potentials in U235 eq
Particulate matter	Particulate matter formation in PM2.5 eq
Human toxicity, cancer and non-cancer	Comparative Toxic Unit for Human Health in CTUh
Ecotoxicity, freshwater	Comparative Toxic Unit for ecosystems in CTUe
Resource depletion – minerals and metals	ADP ultimate reserves in Sb eq
Resource depletion – fossil energy carriers	ADP fossil in MJ
Land use	Land occupation in m ² * a
Water scarcity	Scarcity-adjusted water use in m ³

Pollutant	Acidification	Eutrophication	POCP	Particulate matter formation (PMF)
CO	0	• 0	0.0456	0
NH ₃	1.6	0.35	0	0.64
NO _x	0.5	0.13	1	0.88
PM _{2.5}	0	0	0	1
SO _x	1	0	0.0811	0.54
NMVOC	0	0	1	0.012

Non-methane volatile organic compound

Tables recreated from Determining the environmental impacts of conventional and alternatively fuelled vehicles through LCA, Ricardo Energy and Environment ([Link](#))



Lifecycle Impact Assessment (LCIA)

2020 EU Study Example

Summary of overall lifecycle GWP impacts for Lower Medium Cars for different powertrain type

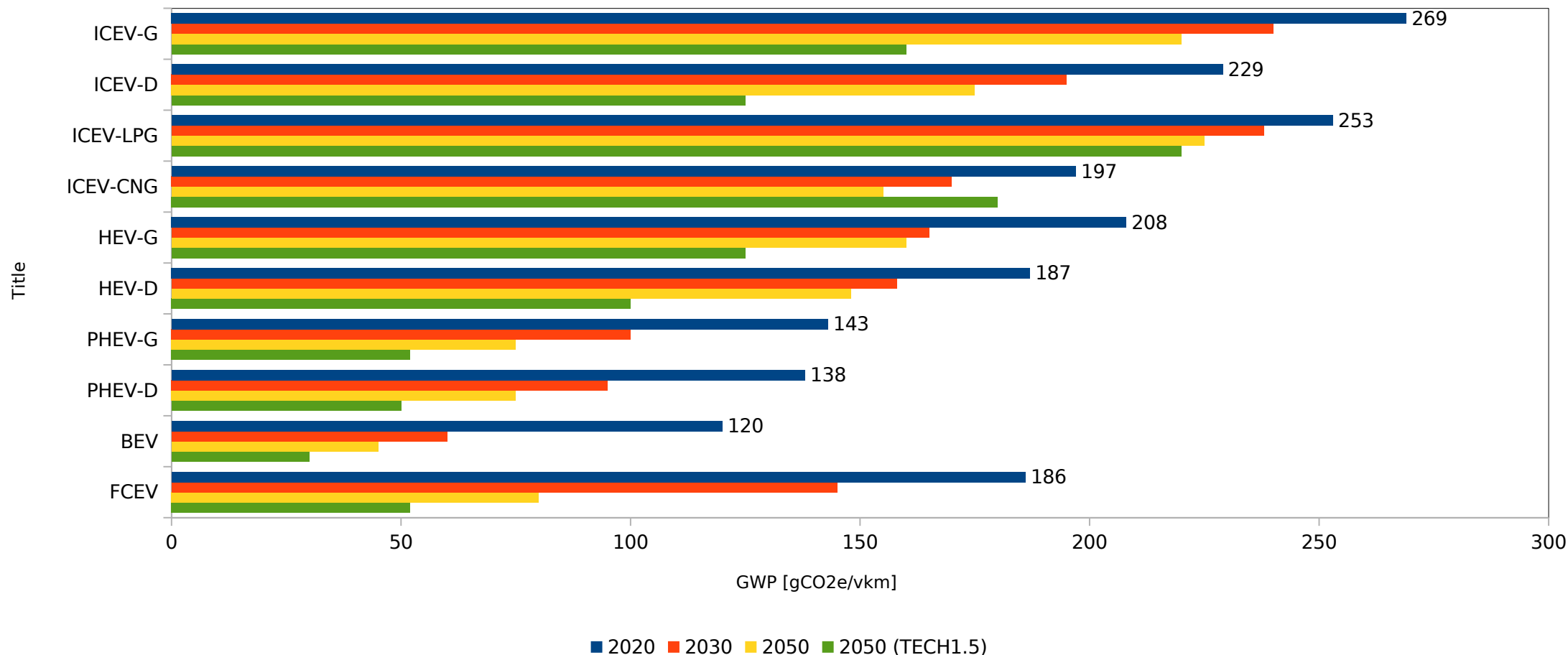


Chart adapted from Determining the environmental impacts of conventional and alternatively fuelled vehicles through LCA, Ricardo Energy and Environment ([Link](#))



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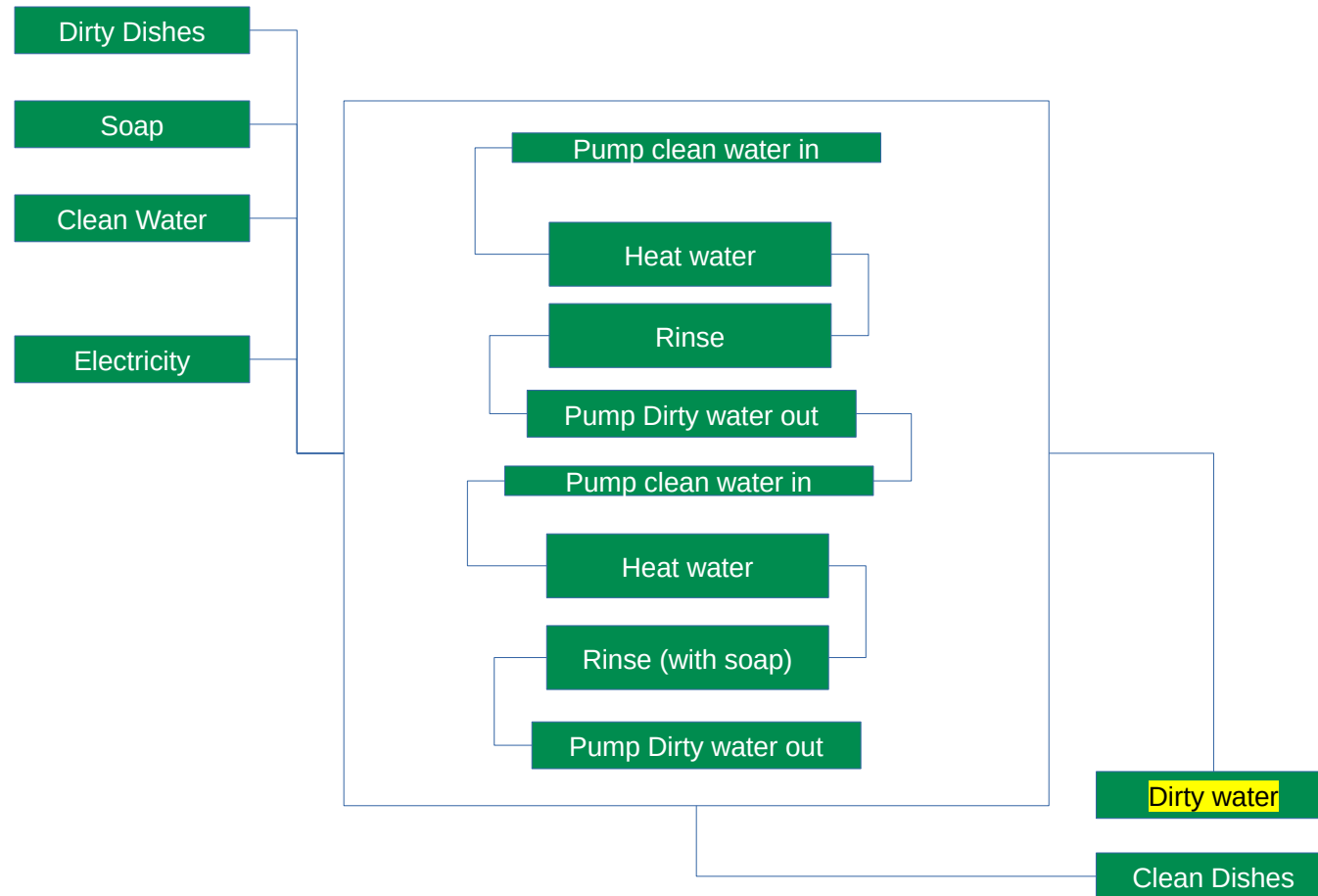
Pollutant	Acidification	Eutrophication	POCP	Particulate matter formation (PMF)
CO	0	• 0	0.0456	0
NH ₃	1.6	0.35	0	0.64
NO _x	0.5	0.13	1	0.88
PM _{2.5}	0	0	0	1
SO _x	1	0	0.0811	0.54
NMVOC	0	0	1	0.012

Non-methane volatile organic compound

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Lifecycle Inventory Analysis (LCI)

Relating data to unit process and functional unit



■ Dishwasher example

— Experimental estimation:

- 1 wash-cycle with 20 dishes, 20 cups and 20 cutlery items used **1.17 kWh** of energy, **12 Litres** of clean water and 30g of soap

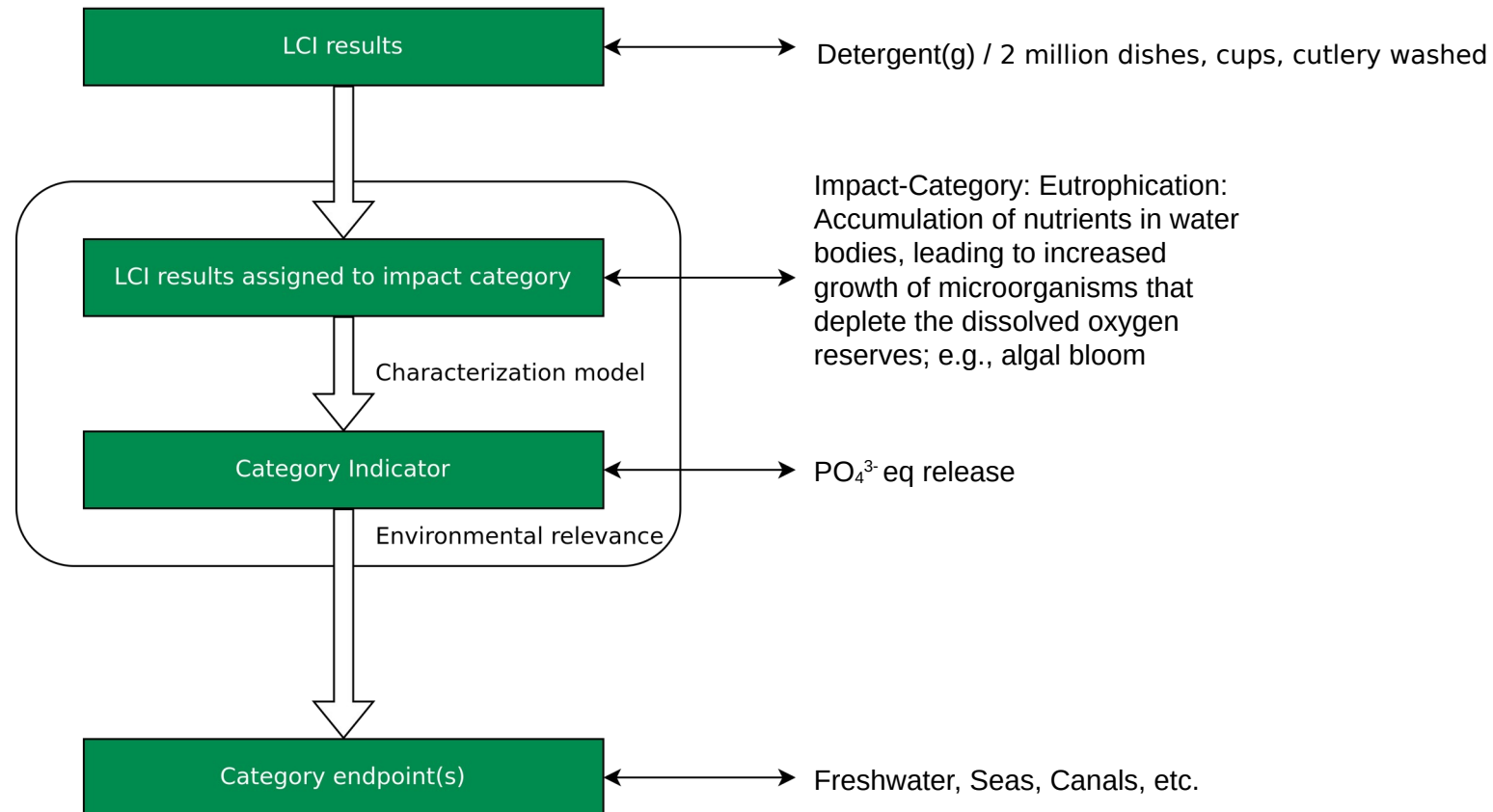
— But the functional unit was: “2 million dishes, cups, cutlery washed”, so we simply multiply the above numbers by **1,000,000**:

- 1.17 GWh, 12 million litres, 30,000 Kg of soap + **200 dishwashers!**

Lifecycle Impact Assessment (LCIA)

Eutrophication from Phosphate-based detergents

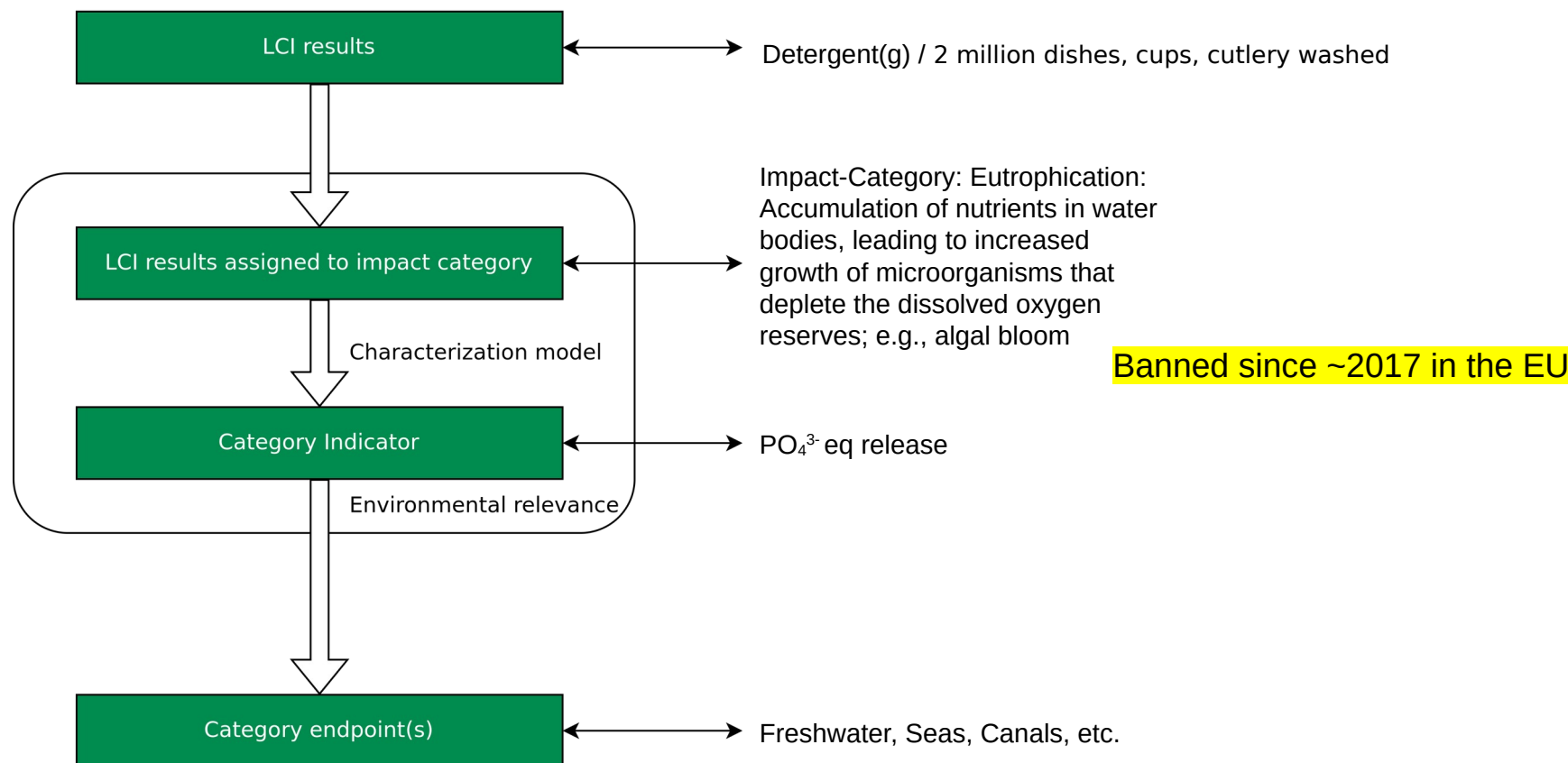
Examples



Lifecycle Impact Assessment (LCIA)

Eutrophication from Phosphate-based detergents

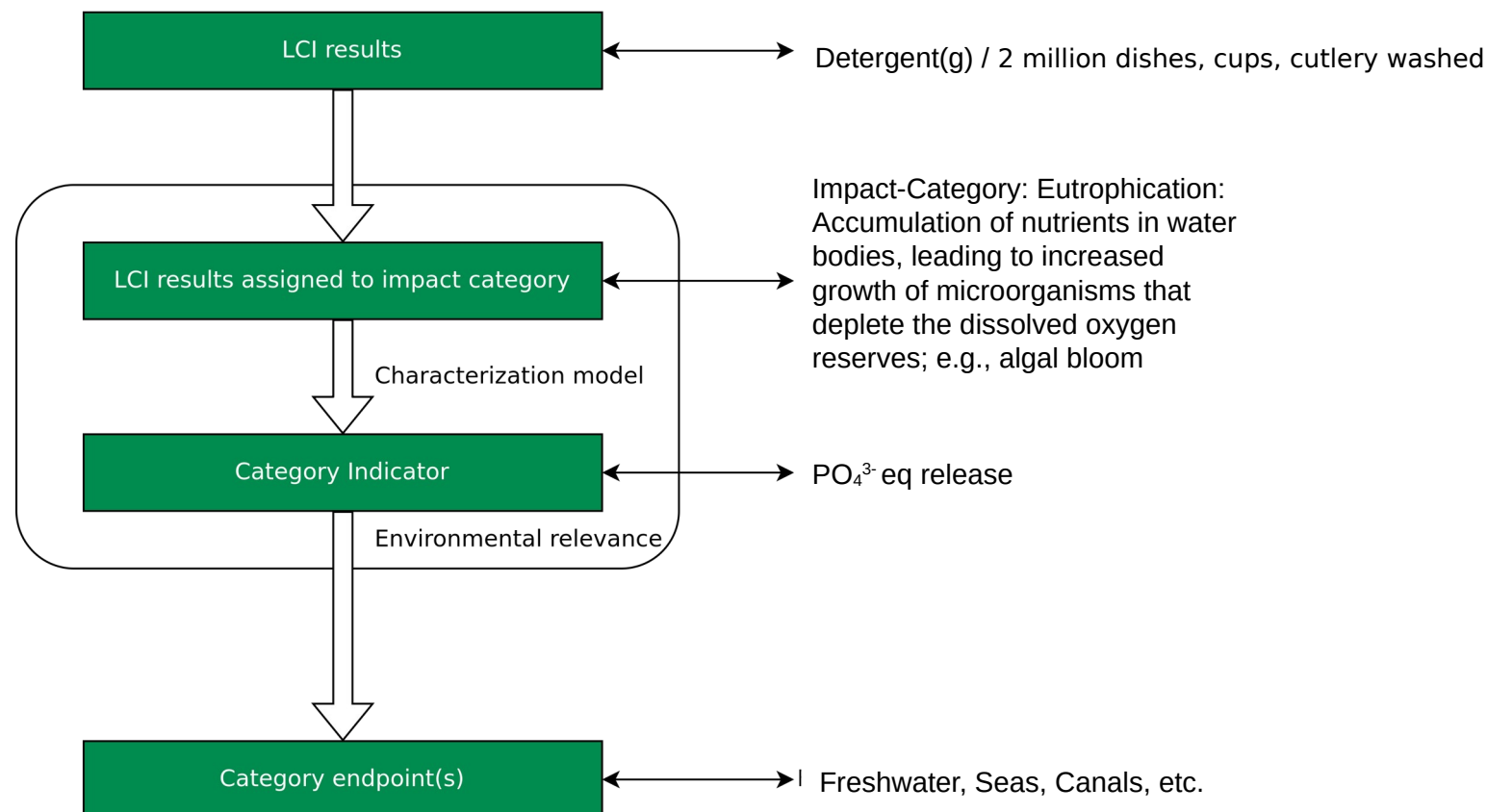
Examples



Lifecycle Impact Assessment (LCIA)

Microplastic pollution from detergent pods

Examples



Lifecycle Interpretation

Definition

Lifecycle Interpretation is the phase of life cycle assessment in which the findings of either the inventory analysis or the impact assessment, or both, are evaluated in relation to the defined goal and scope in order to reach conclusions and recommendations.

ISO 14040 Environmental management — Life cycle assessment — Principles and framework, International standards organisation (<https://www.iso.org/standard/37456.html>)

Determining the environmental impacts of conventional and alternatively fuelled vehicles through LCA, Ricardo Energy and Environment ([Link](#))

Lifecycle Interpretation

Overview

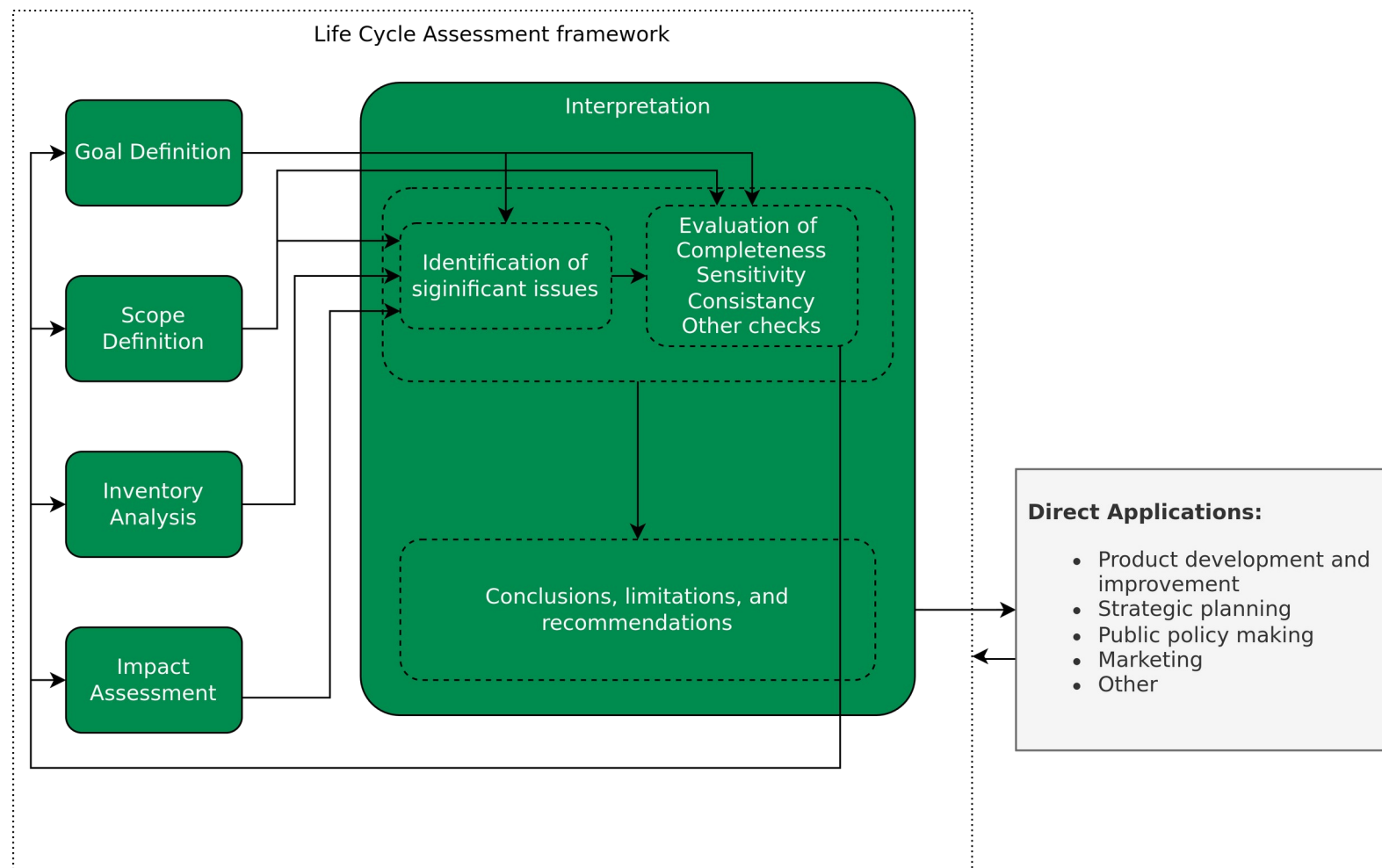


Image adapted from ISO 14040 Environmental management — Life cycle assessment — Principles and framework, International standards organisation (<https://www.iso.org/standard/37456.html>)

Determining the environmental impacts of conventional and alternatively fuelled vehicles through LCA, Ricardo Energy and Environment ([Link](#))

Lifecycle Interpretation

Identification of significant issues

- Two interrelated aspects of significant issues:
 - The main contributors to environmental impacts, like most important lifecycle stages, processes and elementary flows.

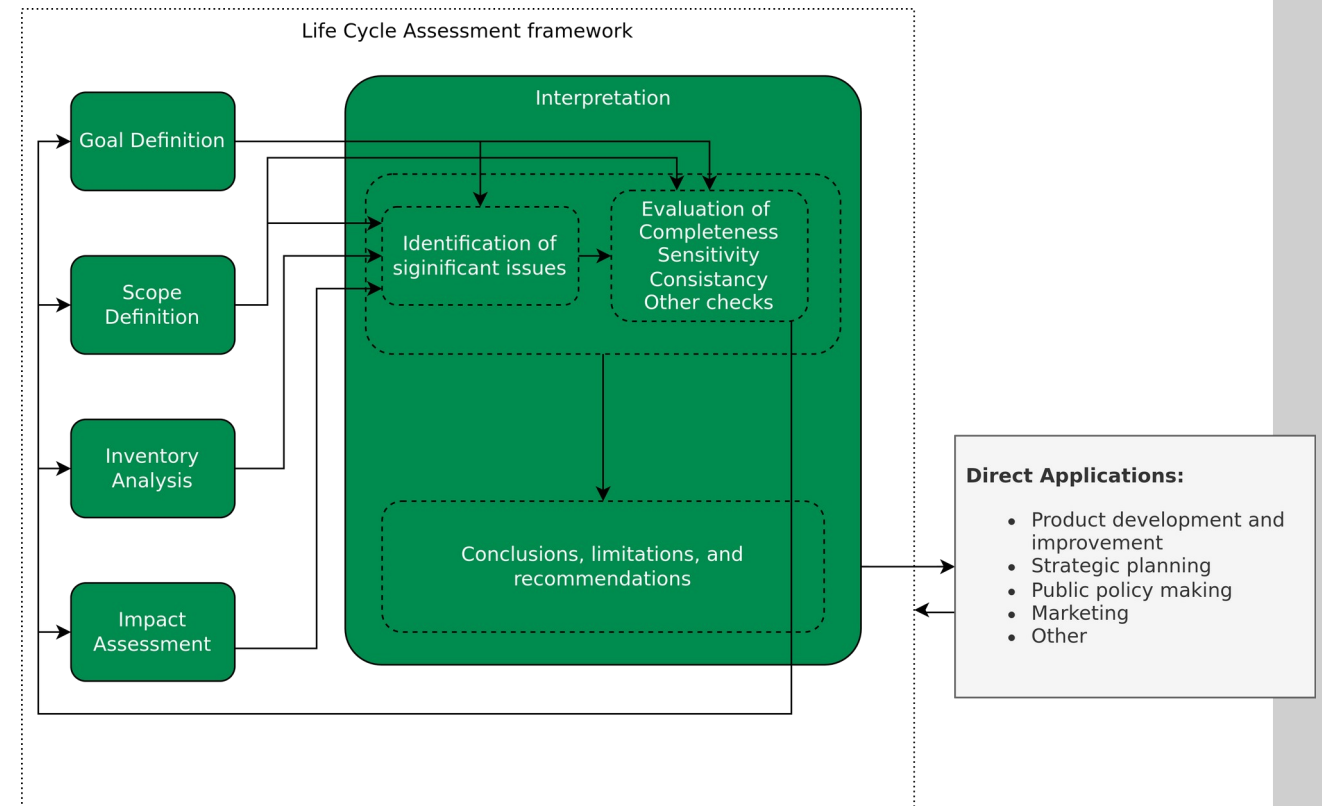


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Lifecycle Interpretation

Identification of significant issues

- Two interrelated aspects of significant issues:
 - The main contributors to environmental impacts, like most important lifecycle stages, processes and elementary flows.
 - The main choices that have the potential to influence the precision of the final results of the LCA, like methodological choices (e.g., cut-offs), assumptions, data, LCIA methods.

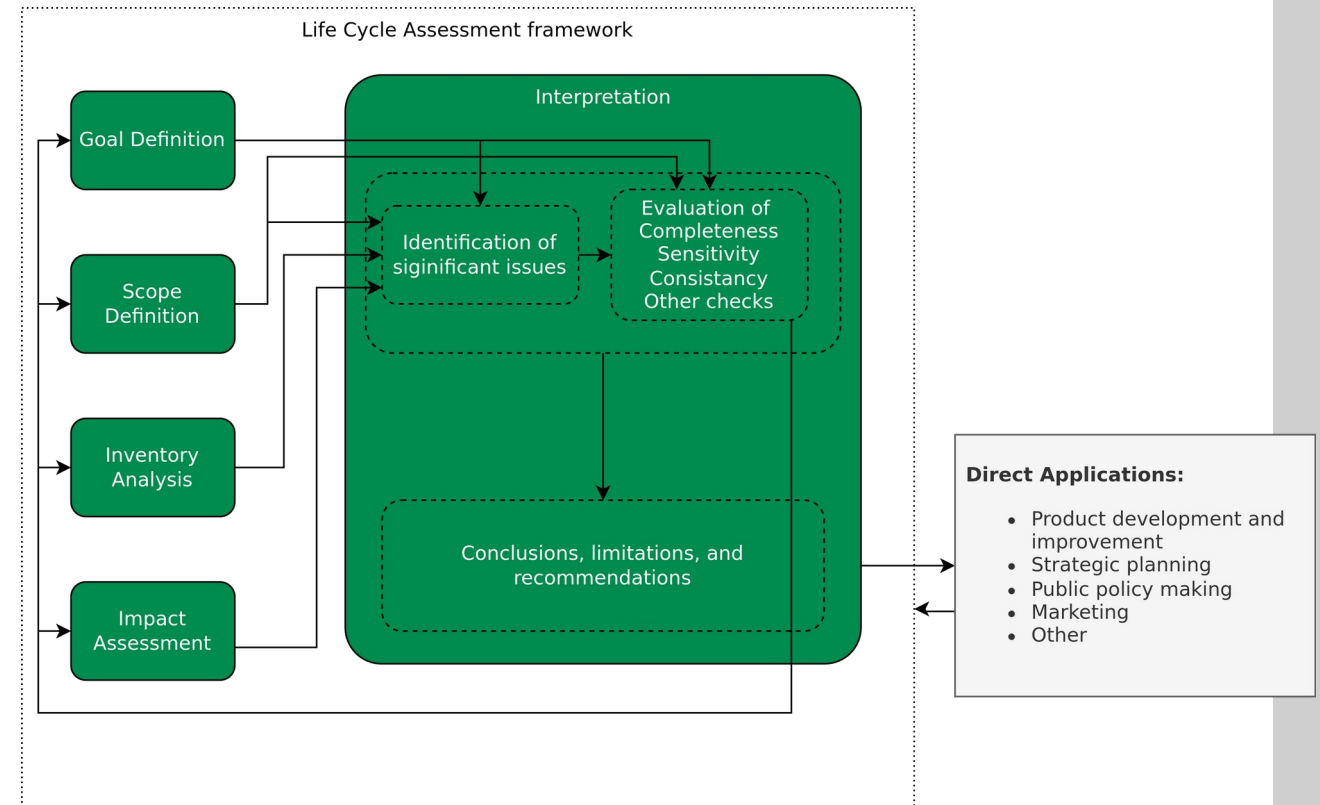
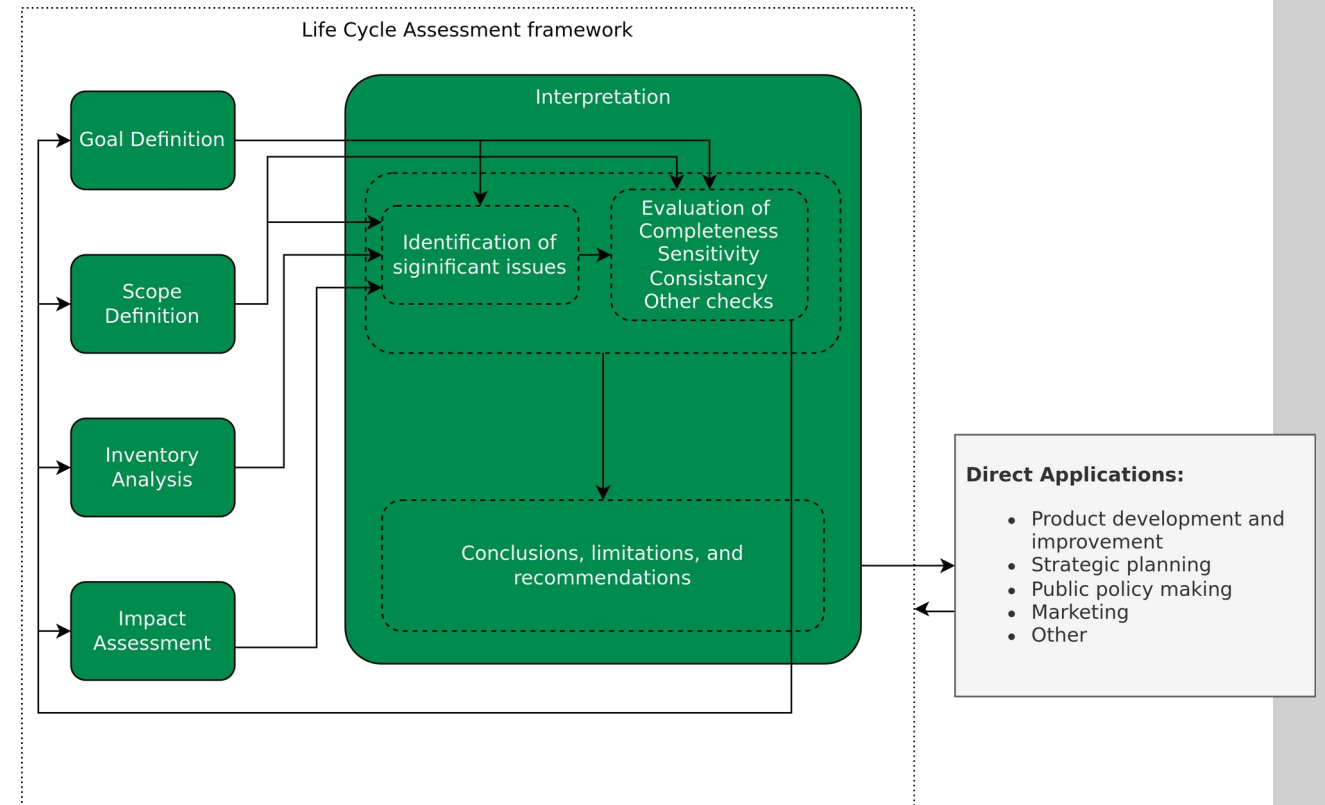


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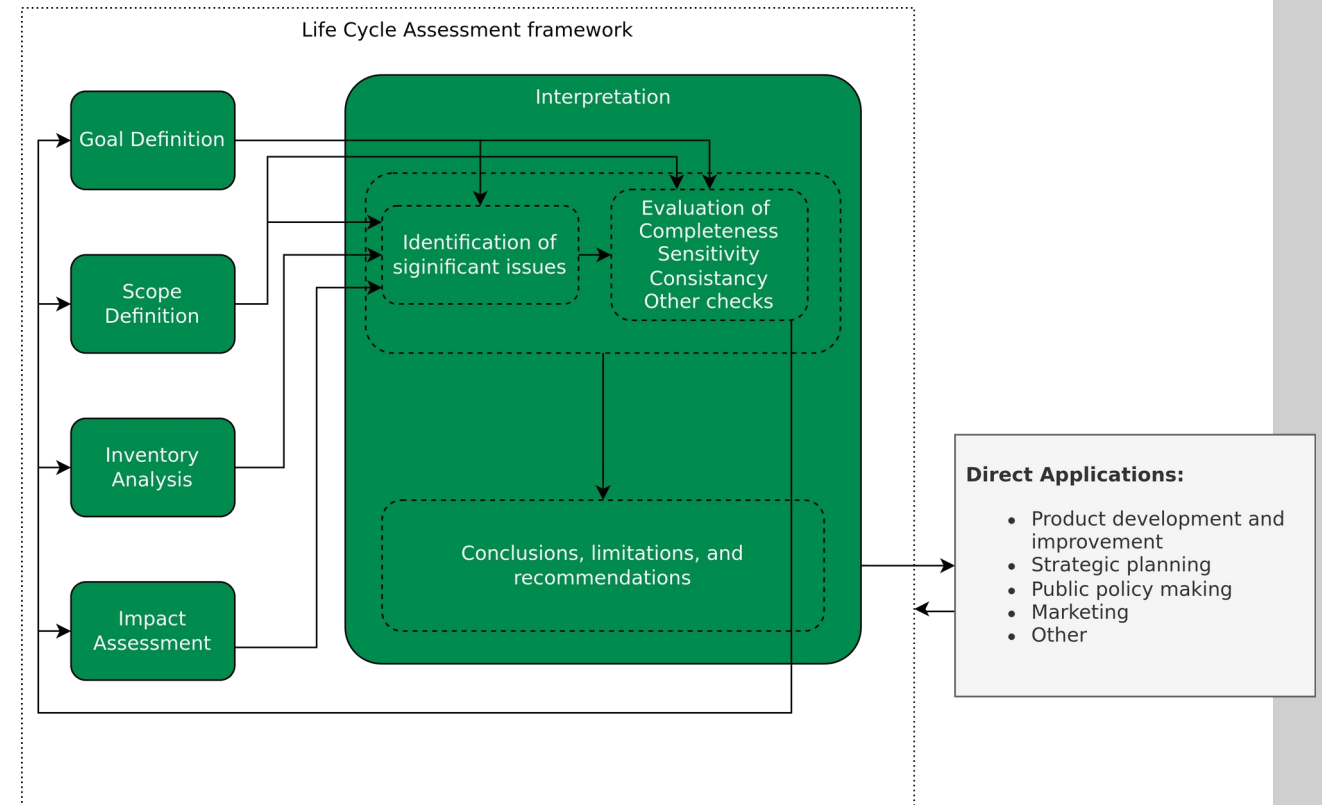
Lifecycle Interpretation Evaluation

- Evaluation is performed to establish the foundation for subsequently drawing the conclusions and provide recommendations during the interpretation of the study results.



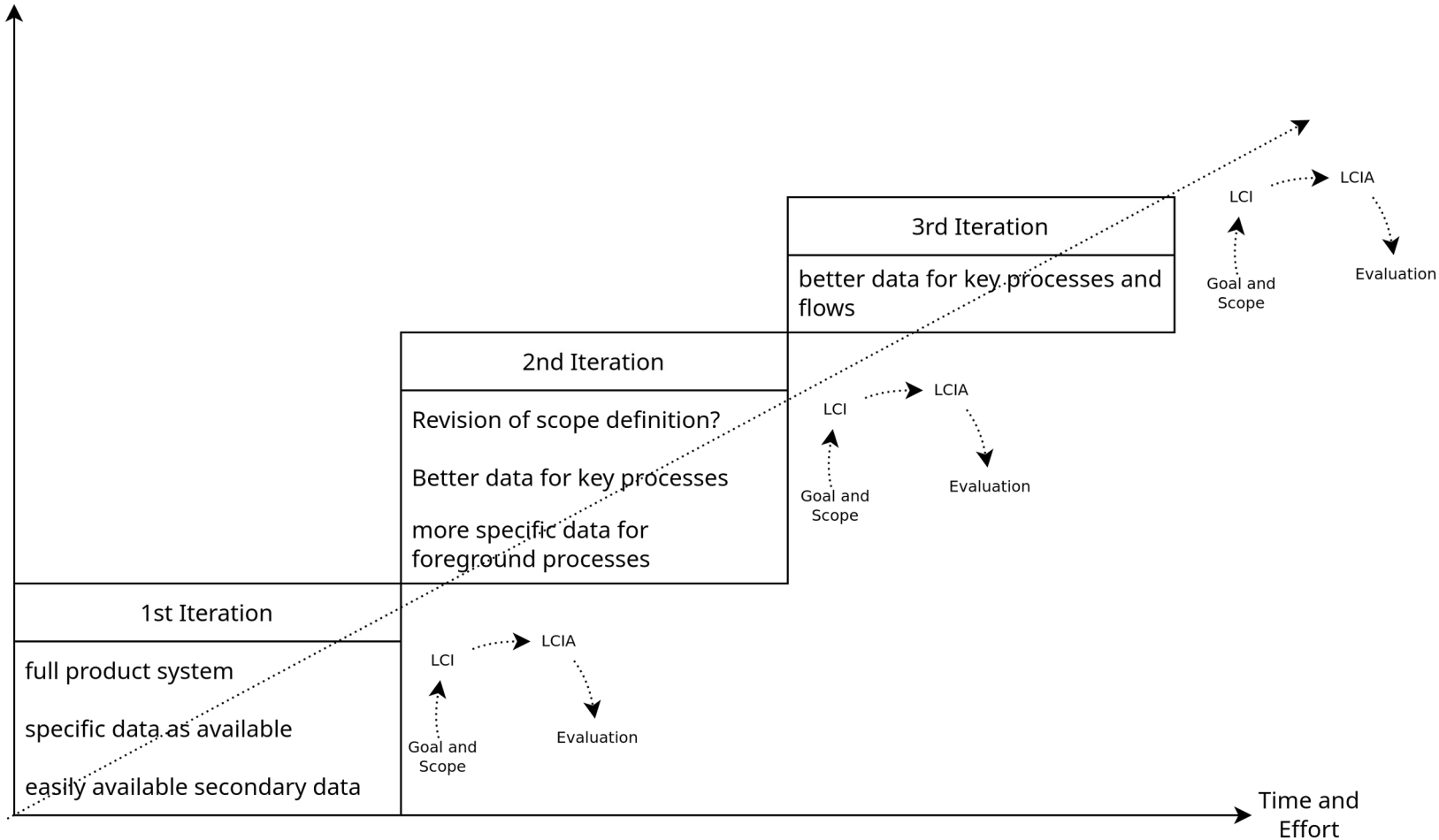
Lifecycle Interpretation Evaluation

- Evaluation is performed to establish the foundation for subsequently drawing the conclusions and provide recommendations during the interpretation of the study results.
- This involves:
 - Completeness checks
 - Sensitivity checks in combination with scenario analysis and potentially uncertainty analysis
 - Consistency checks



Iterative Approach to LCA

Overall data quality
(accuracy, precision,
completeness)



Reporting and Critical Review

- A reporting strategy is an integral part of an LCA.
- A report should:
 - contain the results and conclusions of the LCA in an adequate form to the intended audience
 - address the data, methods and assumptions applied in the study, and the limitations thereof.

Reporting and Critical Review

- A reporting strategy is an integral part of an LCA.
- A report should:
 - contain the results and conclusions of the LCA in an adequate form to the intended audience
 - address the data, methods and assumptions applied in the study, and the limitations thereof.
- A critical review will facilitate understanding and enhance the credibility of the LCA.
- Critical reviews verify whether the LCA has met the requirements for methodology, data, interpretation and reporting and whether it is consistent with its principles.
 - Carried out by an internal or external expert, or by a panel of interested parties.

CONCLUSION

Conclusion

- A high-level overview and guide to Life Cycle Assessment
 - Goal and Scope definition
 - Life Cycle Inventory analysis
 - Unit Processes and Process flows
 - Life Cycle Impact Assessment
 - Impact categories, classification, characterization, weighting, etc.
 - Life Cycle Interpretation
 - Evaluation
 - Reporting and Critical review



EXERCISE 4 – LCA – PART 2

Exercise 4 - LCA - Part 2

- **Main Goal:** Find the environmental impact of using your washing machine
- **Main Task:** Perform LCIA, and Interpret the LCA results.
- **Homework:** Create a detailed list of Impact Categories, such as for Water Use, Climate Change, etc (At least 3 categories), and compute the final impact (category indicator result) for each impact category. Identify the main contributors to these environmental impacts, and recommend a changes that can be made to reduce this impact.

Questions?